REMOTE SENSING IMAGE ACQUISITION, ANALYSIS AND APPLICATIONS

Module One

Acquiring images and understanding how they can be analysed

Voice-over script

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Lecture 1. What is remote sensing?

Slide 1.1.1 The first important concept in this course is to appreciate what we mean by the term remote sensing.

Slide 1.1.2 In simple terms remote sensing is the technology of recording images of portions of the earth's surface from spacecraft, aircraft and other platforms in order to understand the surface by analysing and interpreting those images.

However, image interpretation is not a simple task. So, it is important to understand how the images are acquired in the first instance, and whether those images contain any forms of error or distortion. We start with looking at how energy coming from the earth's surface can be made into an image.

Slide 1.1.3 A remote sensing system consists of a sensor which can detect energy coming from the earth's surface, some form of instrumentation which can convert that measured energy into an electrical signal, and a means for sending that signal to the Earth's surface.

On the ground, the signal is composed into the form of images of the landscape over which the platform has flown. Sometimes the images are formed and recorded on the platform itself and then downloaded when the platform (such as a drone) returns to the surface.

The energy that the sensor detects is most often sunlight reflected from the earth's surface, just as we see the landscape from the window of an aeroplane.

Alternatively, it could be the natural heat energy emanating from the earth, or it could be energy scattered from the surface as a result of some form of artificial illumination. A typical example of this would be scattered radio waves in the case of radar imaging; we will say a little more about this later.

Slide 1.1.4 In simple terms we can look at the remote sensing system in four blocks.

The first is the energy that is detected by the platform. The second is the downlinking of the signal which carries the information recorded by the detector, followed by the subsequent translation of that electrical signal into imagery. This normally occurs in a suitable facility on the ground.

Thirdly, in any real system the images recorded by the platform will have embedded errors. There can be errors in brightness and contrast relative to the actual brightness and contrast of scene and being imaged.

There can also be errors in the geometry of the image compared with the spatial arrangement of features on the earth's surface.

Once those errors have been corrected we have the images in a form that can be used in applications. The images are ready to be interpreted, either by a skilled analyst, or by an analyst with the assistance of suitable computer algorithms. It is this last point which forms the major focus of this course.

Slide 1.1.5 While most of the material to be presented in this course will be concerned with the analysis of images, and thus we might refer to course as a treatment of Image analysis and processing, our principal objective is the science of remote sensing—that is how we can use images from satellites, aircraft and drones to help us understand what is on the landscape below the platform.

We might be interested in the distribution of natural features, we might wish to understand how the landscape is being managed, we might wish to estimate the areas of crops and forests, we might be interested in mineral exploration and geology, or we might want to understand how the landscape evolves with time. These are all objectives of remote sensing, but to achieve those objectives we need to engage in the process of computer-based image interpretation.

So, it is important at all times during our course that we keep in mind that the goal is remote sensing and not image processing for its own sake. To that end we need to understand something about the properties of the surface and the importance of the wavelengths used to form images.

Slide 1.1.6 When thinking about the design of an imaging instrument, our first consideration is what wavelengths the system should be capable of recording.

In principle, we could choose any conceivable wavelength with which to image the Earth's surface. We are guided in this by our own experience.

For example, when we look at the landscape, we view the earth effectively in the three primary colours of blue, green and red (corresponding to the colour receptors in our eyes). They are combined additively to produce the range of colours and brightness that give us the natural set of landscape colours we observe.

But there are of course a whole lot of other wavelengths that we are not capable of seeing. The ultraviolet and infrared ranges are examples. Even though we don't, ourselves, see detail outside the range of human vision, energy does emanate or scatter from the earth over a great range of wavelengths not available to us, but measurable by a satellite sensor.

We will see, shortly, that significant amounts of information about the landscape can the gained if we form images in those other wavelength ranges.

Which are the best wavelengths to use? Before you can answer that we need to understand a little about the atmosphere and how it affects what a sensor can measure.

That is the topic of the next lecture.

Slide 1.1.7 In this slide we have a summary of the important points from this lecture. Each lecture will conclude with such a summary.

Sometimes in these summaries we will add a little more detail and include some points that lead us to think about matters which will be treated in later lectures.

It is important, therefore, not to skip the summaries. Please go back over the lecture as you tick off each dot point in the summary and think carefully about matters that seem to supplement what was presented in the lecture.

Slide 1.1.8 In these simple quizzes we set specific questions that, again, reinforce the lecture just presented but also sometimes ask you to think more widely about matters, many of which are designed as preparation for the next or a later lecture.

Solutions are provided. It would be good if you could work through the questions before consulting the solutions, but the important point is to learn as much as possible from the quiz questions. Therefore, if necessary, consult the solutions!

Lecture 2. The atmosphere

Slide 1.2.1 As we suggested at the end of the previous lecture, we need to understand a little about the atmosphere and how it affects the ability to form images of the landscape. In this lecture we will discuss the properties of the atmosphere and show how it limits the ability to image the earth's surface in certain wavelength ranges while making imaging possible at the other wavelengths.

Slide 1.2.2 The most obvious effect of the atmosphere is that it sits between the source of energy and the earth's surface, and the earth's surface and the sensor. Therefore, the energy from the sun reflecting from a spot on the earth's surface has to travel through the atmospheric column in both directions. We have represented the spot of interest on the ground as a pixel since we will be looking at images in this course.

We will see shortly that the effect of the atmosphere on the transmission of radiation is wavelength dependent. More importantly, we need to understand whether radiation can pass through the atmosphere at all.

Slide 1.2.3 This is quite a complicated slide and we will spend a bit of time on it. At the top it shows the electromagnetic spectrum from the ultraviolet through to long radio waves. In the infrared range it is separated into reflective and thermal infrared. The reasons for this will be seen later. We have also identified the terahertz and millimetre wave ranges separately.

Note that together we group the visible and reflective infrared wavelength ranges under the single title of optical wavelengths.

On the bottom half of the side we show the transmission of the atmosphere as a function of wavelength. As seen, this is a very complex graph and the only wavelengths for which the atmosphere seems to be totally transparent (that is it has a transmittance or transmissivity of 100%) is in the range of radio waves. That is very reassuring because it means that a clear atmosphere does not affect radio, television and mobile (cell) phones.

In the terahertz and millimetre wave range the transmittance of the atmosphere is almost zero, which means those waves are almost entirely blocked by the atmosphere.

In the visible and infrared ranges, while the atmosphere is not entirely transparent, it does allow significant transmission of electromagnetic radiation. There are certain small selected bands where is radiation is blocked but there are plenty of wavelengths where good transmission is possible.

The reasons for the atmosphere absorbing radiation at certain wavelengths are several and complex. At the shorter wavelengths atmospheric ozone is a strong absorber. At most other wavelengths water vapour and carbon dioxide selectively absorb radiation as indicated in the diagram. At the long radio wavelengths, the layer in the upper atmosphere called the ionosphere significantly refracts radiation so that those frequencies are also unusable for imaging purposes.

The wavelength ranges (large or small), over which the atmosphere shows good transmittance, are called atmospheric windows.

Slide 1.2.4 As we noted on the previous slide, the principal atmospheric constituents which absorb radiation in the ranges of interest to remote sensing imaging are water vapour and carbon dioxide. This slide shows graphs of the transmissivity (transmittance) those two constituents over the visible and infrared, along with the transmissivities of oxygen and ozone. As seen, the latter are much less important than water vapour and carbon dioxide at those wavelengths.

Slide 1.2.5 By way of summary, we can conclude that the most suitable ranges of wavelengths for imaging the earth's surface through its atmosphere are the visible and lower end of the reflective infrared range (called near infrared). As a result, if we examine the optical sensors used on most remote sensing satellites, we see they tend to have imaging capabilities somewhere within that broad range.

However, we will see that there are other wavelengths that can be used if we re-examine the transmission properties of the atmosphere over the full wavelength range.

Slide 1.2.6 Returning to our previous graphs we see that there is a reasonably good atmospheric window in the thermal infrared (i.e. heat) range, and exceptionally good transmission for radio waves.

Thermal remote sensing satellites, i.e. those that detect heat emanating from the earth's surface, operate with wavelengths in the range of about 3-5mm. Those satellites can form heat maps of the surface, and are particularly good for detecting burning bush and wildfires.

Because there is very little natural radio energy emanating from the earth itself, it is usual to irradiate the surface with an artificial source of energy if we wish to take advantage of the very broad atmospheric window in the radio wave range. The remote sensing platform carries a radio receiver which is so configured that it forms an image of the surface at radio wavelengths. That is the basis of the very large field of radar remote sensing.

Sometimes we do form images of the natural microwave emissions of the surface, but because the energy levels are so low, they tend to have very poor resolution.

While we won't be looking at the technology of thermal remote sensing we will look radar remote sensing in last half of module 3. The image analysis techniques we will treat are generally applicable to those two imaging modalities as much as they are to the interpretation of optical remote sensing images.

Slide 1.2.7 The important message from this lecture is that the atmosphere interferes selectively with the passage of electromagnetic radiation. When designing a remote sensing program attention must be paid to those portions of the spectrum which allow reasonable transmission of radiation.

Slide 1.2.8 In addition to the set questions, you might like to think about how you would design a remote sensing imager to map the surfaces of Mars and Venus. That requires you to think a bit about the atmospheres of those planets and the particular challenge of seeing the surface of Venus.

Lecture 3. What platforms are used for imaging the earth's surface?

Slide 1.3.1 We now look at the sorts of platforms that are routinely used for gathering remote sensing imagery. Over the past 40 years or so, the most common platform has been the earth orbiting satellite. Nevertheless, aircraft platforms are often used and, more recently, imaging from drones has become popular. In this lecture we will say a little about each platform type, concentrating on satellites since their orbital characteristics are quite specific to the need for operationally monitoring the globe.

Slide 1.3.2 There are several differences among spacecraft, aircraft and drones for imaging, many of which we will explore in later slides. The most obvious though is that the further one is from the earth the greater the area that can the imaged, whereas, based on our own experience, higher spatial resolution would seem to be possible with platforms closer to the surface. That immediately sets satellites apart from the other two; by using spacecraft platforms it is technically feasible to image the whole of the Earth's surface in a practical timeframe.

Aircraft and drone missions, by comparison, are not global imagers, but tend to be focused on mission-specific areas of interest.

Slide 1.3.3 We will now look at each of the three platforms in a little more detail.

As listed here, satellites will allow the whole earth globe to be imaged within a reasonable timeframe. Also, importantly, they operate above the earth's atmosphere. While that means imaging must be carried out through the atmospheric column, as discussed in the previous lecture, they benefit from not travelling *through* the atmosphere, meaning the platform is quite stable. The atmosphere can often be turbulent, meaning that platforms that travel through it can have random variations in their altitudes and attitudes, causing variations in the geometry of recorded images.

Nevertheless, imaging through the atmospheric column will introduce brightness and contrast errors into the recorded image data which often requires correction before the image is usable.

Because satellite programs are so expensive to develop, build and launch, the data tends to be made available to many thousands of users and thus tends to be regarded as a common-good product, notwithstanding the fact that some countries restrict access to the images recorded by their national programs. The same of course is true of commercial satellite operators; the user base is very large, even though the products are sold at commercial rates.

Two final points should be noted. Because satellites are placed in orbit and thus are not easily re-visited, the sensor characteristics are fixed by design when the program was under development. As a consequence, the imaging modality cannot easily be reconfigured.

Secondly, there is a move now to clusters of microsatellites for imaging purposes. This approach benefits from the much lower cost of the platforms and, with time, will also benefit from collaborative decision-making among platforms. That is still a little way off.

Slide 1.3.4 When aircraft are used to carry imaging sensors a number of benefits are immediately apparent, including

- the ability sometimes to choose the imaging wavelengths which best match the application of interest
- the ability to achieve high spatial resolutions, and
- the control given over the region to be imaged and how frequently imaging should take place.

On the negative side, atmospheric influences, including turbulence, affect the stability of the platform and therefore tend, on the average, to introduce more geometric distortion than would be the case with satellites.

However, since the atmospheric column is generally small, distortions in brightness and contrast tend not to be as significant as with satellites.

Although satellite missions are expensive to launch and maintain, they have an economic benefit in that the imagery is available to a large number of users. For aircraft missions the end user generally has to pay for the full mission costs, so that aircraft imaging is an expensive option if satellite-based imagery is a viable alternative.

Slide 1.3.5 We still have much to learn about the use of unpiloted vehicles such as drones for serious remote sensing purposes. Most of the characteristics of aircraft imaging apply to drones as well, apart from cost. In general, the use of quadcopters, and the like, is a fairly inexpensive imaging modality.

Although some concerns have been expressed over privacy with aircraft imaging, most of the time that is not a problem because of the altitude at which the platform flies. With drones, however, privacy is a major consideration particularly in urban regions, because of the low flying altitudes.

Slide 1.3.6 We now need to understand a bit more about the orbital characteristics of satellites when used as remote sensing imaging platforms. Of particular importance is how we can arrange a satellite in orbit so that it can be used to image all of the earth’s surface in a practical timeframe and do so repetitively.

Slide 1.3.7 First, without going into the details of orbital mechanics, it is sufficient to notice that most remote sensing satellites orbit quite close to the surface—typically between 600 and 900 km. That is referred to as a low earth orbit (LEO). In such an orbit the satellite takes about 90 minutes to do one complete revolution.

And here is a neat trick which is fundamental to almost all operational satellite remote sensing missions: the orbits are arranged to be near-polar. As such, the earth is rotating eastwards underneath the satellite and on each adjacent orbit the satellite is travelling over a different portion of the earth's surface. The orbital inclination is arranged so that it is possible to have adjoining strips of images on each orbital pass.

Also, on each pass over, say, the equator the satellite sees the same local (sun) time. Such an orbit is called sun-synchronous. If the orbit crosses over the equator during the daytime, travelling from north to south, that crossing is called a descending node.

A descending node of about mid-morning is chosen for most programs, so that there is a good level of reflected sunlight; but there is also sufficient shadowing to make topographic detail visible.

The sensors carried image directly underneath the satellite so that the forward motion of the platform allows a strip (or swath) of imagery to be recorded.

In order to achieve global imaging, the orbits are arranged to repeat on a given cycle, called the repeat cycle. Typical examples are 16 days for Landsat 7 and Terra and 26 days for Pleiades and Spot.

Slide 1.3.8 This slide shows a sun-synchronous orbit in a little more detail, this time for a satellite in an ascending node configuration.

As seen, the first orbit comes up over the Indian Ocean, the next (second) over Central Africa and the third over the Atlantic Ocean just off the African coast.

It is not the orbit of the satellite itself the changes but the rotation of the earth underneath the orbit.

Slide 1.3.9 This summary effectively is a statement of the important characteristics of the orbit of a remote sensing satellite.

Slide 1.3.10 The third question in this set asks you to think in each case about whether a satellite, aircraft or drone would be the most appropriate imaging platform.

Lecture 4. How do we record images of the earth's surface?

Slide 1.4.1 In the last lecture we looked at the types of platform that are used to carry remote sensing imaging instruments. We now wish to look at the instrument technologies commonly employed. That is the purpose of this lecture.

Slide 1.4.2 Earth imaging could, of course, be carried out by mounting a simple digital camera on a platform. Today, that is often done with drones.

Many of the earlier images of the earth taken by astronauts were captured by film cameras. And the first three of the Landsat satellites used a television camera (called a Return Beam Vidicon) as one of its instruments.

Most remote sensing satellites carry scanners of one form or another to capture images of the earth's surface. They tend to make use of the forward motion of the platform, along with some form of scanning, either across or along the path of the satellite, in order to build up an image of the landscape.

In this lecture we will look at the three most common types of image scanner.

Slide 1.4.3 The earliest digital imager was a line scanner, a technology which is still in use. It uses a mirror to scan across the path of the satellite, as shown in this side, in order to focus strips of the earth's surface onto a detector carried on the platform. By design, the rate at which the mirror scans and the forward velocity of the satellite are arranged so that consecutive scan lines are adjacent to each other and together form an image as the satellite traces out its path.

The mirror has to oscillate from side to side. In sensors like the Landsat Multispectral Scanner (MSS), upwelling radiation from the landscape is detected on the forward scan of the mirror. The mirror then resets to the opposite side of the satellite track ready for the next scan. By contrast, the Landsat Thematic Mapper, and its later variants, records data on both the forward and reverse scans. Sometimes, particularly in early aircraft sensors, a rotating mirror is used.

This slide also introduces some important definitions. First, the width of the image recorded is called the swath. It is defined by the Field of View (FOV) of the sensor, which is the total angle over which it scans.

The resolution of the sensor, or scanner, is called the Instantaneous Field of View (IFOV). Again, this is defined by an angle which, when projected onto the Earth's surface, defines the pixel size in, or spatial resolution of, the recorded image. Most often, the IFOV is the same in the along-track and cross-track directions.

Slide 1.4.4 The mechanism described in the previous slide accounts for the scanning process. But most remote sensing imaging instruments are capable of recording in several different wavebands simultaneously. With the line scanner that is achieved by placing a dispersive element in the path of the radiation. This can be a diffraction grating or prism, the effect of which is to break up the incoming beam into individual beams corresponding to the wavelengths of interest. Those beams are then focused onto separate detectors–one for each waveband.

Slide 1.4.5 Another common means for forming an image is to use a linear array of individual detectors, arranged across the direction of satellite motion. Usually this is a CCD (charge coupled device) array with one element for each of a series of longitudinal (i.e. along the track of the satellite) scan lines. Radiation falling on each detector is sampled in turn allowing pixels to be formed. The pixel size is defined by the detector's instantaneous field of view at the surface and the rate at which the radiation is sampled compared with the forward velocity of the platform.

No mechanical scanning is involved apart from the forward motion of the satellite. The advantage of this technology is that more radiation can be gathered per effective pixel allowing higher spatial resolutions to be achieved.

Again, a dispersive device is incorporated to allow the simultaneous recording of several wavebands. There will be a separate detector CCD array for each waveband.

Slide 1.4.6 Perhaps the most recent imaging technology used on spacecraft and aircraft platforms is similar to the push broom concept just discussed but, instead of having a separate set of linear CCD arrays to record a discrete set of different wavebands, a two-dimensional CCD array is employed. The second dimension is used to replicate the effect of a very large number of separate linear CCD arrays. In this manner the number of wavebands recorded simultaneously can be very large, typically up to 200 or so. Therefore, these sensors typically record simultaneously several hundred co-registered images of the landscape ranging over the visible and infrared wavelengths.

As we will see later that will allow a great deal of information to be obtained about the landscape being imaged.

When the number of recorded wavebands is small, say 20 or lower as for the two previous instruments, the devices are typically called *multispectral* scanners. When the number of wavebands can be in the order of 100 or so, as in this slide the instruments are usually called *hyperspectral* scanners.

Slide 1.4.7 Now that we have seen how images can be recorded, we can summarise their important properties when used for remote sensing purposes.

Recall, first, the geometric properties of the scanner shown on the right-hand side of this slide. The IFOV of the scanner defines the spatial resolution (or pixel size) on the ground, whereas the FOV of the scanner defines the swath width. Also shown in the diagram are the typical units use to describe each of those properties in practice.

On the left-hand side of the slide we have summarised how an image in remote sensing is described. Overall it is defined by the swath width in the cross-track direction and the frame size in the along-track direction. Most often, particularly for operational missions, the frame size is chosen to be the same as the swath width.

The next important feature is the pixel size, which determines the level of detail about the earth's surface being recorded. As indicated previously we often refer to pixel size as the spatial resolution of the image.

Clearly, we are also interested in the number of wavebands being recorded. As indicated here, the wavebands are all registered with each other geometrically.

Finally, we need to know the number of brightness levels available for each pixel. For a black and white image those levels will be gradations from black to white. In fact, any image recorded in a given waveband will effectively be a black and white image, representing levels of received radiation from zero to the maximum sensitivity of the detector. Later on, we will see how we display those recorded bands in colour.

The number of brightness values available per pixel is defined by binary arithmetic. For example, an 8-bit pixel is capable of representing 256 levels of grey. Most current operational remote sensing images will have 8, 10 or 12 bit pixels. The number of brightness values (often expressed as the number of bits) is called the dynamic range or radiometric resolution of the sensor.

Slide 1.4.8 Although it is of interest to see how line scanners and hyperspectral Instruments operate, the really important message from this lecture is the geometric characteristics of the sensors (IFOV, FOV etc.) and the properties of the image itself.

Slide 1.4.9 The questions here ask you to think a little more laterally about how images are recorded. In particular, the third question raises an important matter about the distortions in geometry that can occur for imaging systems with wide fields of view.

Lecture 5. What are we trying to measure?

Slide 1.5.1 In this lecture we want to look at the properties of the earth's surface that can be detected by remote sensing instrumentation. When we examine those properties we will understand why certain wave bands have been chosen in the design of the sensors commonly-used in civilian remote sensing missions.

Once you have assimilated this material you may wish to consult some of the many websites available that include information on remote sensing programs and their sensors. As we develop further knowledge in later lectures, these websites will become valuable additional sources of information with which to illustrate the lecture presentations. The NASA website, in particular, is especially valuable.

Slide 1.5.2 First, remember that most remote sensing missions record images of the earth's surface using reflected sunlight, just as you and I see the earth as a result of reflected sunlight. In remote sensing, though, wavelengths outside the range of human vision can also be used for imaging. But let's just concentrate on reflected sunlight.

If the earth's surface was an ideal reflector, such as a sheet of white material, then all wavelengths would be reflected equally.

Real earth surface materials, however, are not ideal reflectors. They selectively absorb sunlight at different wavelengths, giving the appearance of colour in the visible range. For example, green vegetation appears to us as green because in the visible range of wavelengths there is strong absorption of incident blue and red radiation from the sun, but less absorption of green sunlight. As a result, there is more green energy reflected from most plant matter, so that we see grass, trees and other green vegetative matter as green.

Depending on their pigmentation, some plants are differently coloured. For example, plants which appear red strongly absorb incident blue and green radiation but reflect a significant proportion of the incident red radiation from the sun.

To develop a better idea of what satellite sensors detect, including in the ranges of wavelengths beyond human vision, we will now examine in a little detail the reflectance properties of earth surface materials as a function of wavelength.

Slide 1.5.3 This slide shows how the three simple fundamental earth cover types of vegetation, soil and water reflect incident sunlight across just the ***optical*** range of wavelengths.

Note that the vertical axis is calibrated in percentage terms and is expressed as "reflectance".

Note, also, that the range of visible wavelengths, shown at the left-hand end of the abscissa, is quite small compared with the full range of optical wavelengths.

It is important to note that the amount of green energy reflected by (green) vegetation is really quite small. Nevertheless, it is larger than the blue and red reflectances, meaning we see most vegetation as green, even though the energy levels are quite small. By contrast, in the near infrared range, vegetation reflects about 30% or more of the incident sunlight. If we were able to see in the near infrared, vegetation would look much brighter than it does in the optical range.

As we move to longer wavelengths still, we see that the vegetation reflectance curve drops away. It contains two large absorption dips resulting from the water content of the vegetative matter. Depending upon the moisture content of the vegetation those dips can be very deep or quite shallow. If the vegetation is moisture stressed the dips will be shallow to non-existent. Most remote sensing instrumentation tends to avoid imaging in those regions.

The water absorption dips also occur for soil, although in this real example the soil is quite dry so that the dips are very shallow. Soil with a high moisture content will show dips comparable to those in this case to the vegetation curve.

The other interesting property about soil is that the reflectance curve rises steadily with increasing wavelength in the visible and near infrared ranges. As a result, the visible colour of soil is determined mostly by reflected green and red radiation, so that the colour tends towards reddish-yellow. That is particularly the case for clays and sand. Dark loamy soil will appear black because there is high absorption at all wavelengths.

The water reflection curve is dark everywhere showing just small amounts of reflected energy in the blue-green range consistent with our experience of water colour.

These three curves are typical of what we encounter in remote sensing, but they will vary considerably depending upon the moisture and mineral contents of soil, the moisture and pigmentation contents of vegetation and the clarity and depth of water. If water contains a lot of sediment, for example, the water curve will be a combination of the reflectance of pure water and that of the suspended soil matter. If the water is shallow, there can be reflectance from the bottom material. So, when looking at cover types such as water, the remote sensing specialist needs to be aware that the reflectances observed can be mixtures of more pure materials.

Slide 1.5.4 In this slide we show how the wave bands of some common remote sensing satellite instruments are chosen specifically so that they sample the most significant regions of the spectral reflectance curves of common earth surface cover types.

To consider a simple example, the Landsat ETM+ takes three measurements in the visible range, roughly corresponding to blue, green and red; another measurement just at the start of the near infrared range; and two broad measurements further out in the infrared, avoiding the water absolution dips.

Sensors, such as Hyperion, take a very large number of samples across the spectrum, with sufficiently fine spectral resolution that, in principle, the full reflectance spectra could be reconstructed for each pixel. Remember, each pixel is sampled in all of the available wave bands so that the spectral information recorded by any of these sensors is done for every pixel in the image.

Whenever you come across a sensor have a look at the wave bands it operates with and form an opinion as to its likely application. If a sensor operates largely in the visible range it is probably designed for surveillance, mapping or urban studies. If it has wave bands in the near infrared it is then good for mapping and monitoring cover types such as natural vegetation, crops and forests. If it operates further out in the infrared range it is likely to be designed for geological studies as well, since a number of important spectral features of minerals occur in the wavelengths range up to 2.5µm.

Slide 1.5.5 We are now in the position to introduce some further definitions.

If an instrument records up to about 10 wave bands it is generally referred to as a ***multispectral sensor***. If it records a very large number of bands, say more than 100, it is often referred to as a ***hyperspectral sensor***.

Slide 1.5.6 At this stage it is worthwhile re-examining other portions of the electromagnetic spectrum, and in particular the region beyond the optical wavelengths. Remember, though, we have to be mindful of the presence of atmospheric absorption and can only reliably image over those wavelengths which correspond to atmospheric windows.

As we noted in a previous lecture there are good atmospheric windows in the vicinity of 10µm, and in the very broad radio wave region of the spectrum. They are identified on this slide and we need to ask ourselves, once again, whether those regions are useful for earth surface imaging.

Let’s look at the wavelengths in the vicinity of 10µm first.

Slide 1.5.7 There is a very famous law, called Planck's radiation law, which describes how a body—a so-called blackbody—radiates energy as a function of wavelength. The radiation emitted is also, importantly, a function of the temperature of the body.

Planck's law is described by an equation that allows us to plot radiation curves as a function of wavelength and temperature. In this slide we show three examples of Planck's radiation law in practice. The first example is for the sun, whose surface temperature is taken to be 5950K. The second is for the earth, which has a temperature of approximately 300K. The third is for an object with a temperature of 1000K, which roughly corresponds to a burning fire.

As seen, they have maximum radiation outputs at different wavelengths. It is important to note that the vertical scale is logarithmic, so that each integer spacing represents a power of 10 (or an order of magnitude). Also, the sun curve has been scaled down to represent the reduction in radiation because of the distance it has travelled from the sun to the earth.

The solar curve has a radiation maximum at a wavelength of about 0.5µm, whereas the earth's maximum emission is at a wavelength of about 10µm. The two curves cross over a little below 5µm, meaning that any observations of the earth's surface at wavelengths less than about 4µm are dominated by reflected sunlight, whereas emissions above about 6µm are dominated by natural emissions from the earth itself.

If we want to measure the natural energy emanating from the earth, we choose a sensor with wave bands in the range of 10-12µm, near where the earth radiation is at a maximum. That is the basis of, so-called, thermal remote sensing, since we are effectively detecting variations in the earth's heat output.

Although we have shown the earth curve as a simple continuous line, it does vary from place to place on the earth as a result of a property called emissivity. It is variations in emissivity that thermal remote sensing missions measure and map.

Slide 1.5.8 If we extrapolated both the earth and solar curves of the previous side out to the radio wavelengths, we would see that there is negligible emitted or scattered energy from the earth's surface at those very long wavelengths.

In fact, there is some and it can be detected with very low spatial resolution instruments, but in general we assume that the earth is quiet at radio wavelengths. As a result, we can irradiate the surface using an artificial energy source, such as a radio transmitter carried on an aircraft or spacecraft. A receiver on the platform detects scattered radiation which is then used to create a radio or microwave map of the earth's surface. That is the basis of radar remote sensing.

Slide 1.5.9 There are three important messages here:

1. Imaging can be done with reflected sunlight, thermal emissions from the earth itself, or via scattering of radio waves (radar)
2. Note the difference between hyperspectral and multispectral imaging
3. Water absorption bands need to be avoided

Slide 1.5.10 When thinking about the first question, ask yourself how a burning fire front could be mapped against significant surface features such as roads, rivers and forests.

Lecture 6. Distortions in recorded images

Slide 1.6.1 Images are never perfect when recorded.

It is the purpose of this lecture, and those which follow, to understand that the images acquired by remote sensing missions contain errors compared with the region on the ground being imaged.

Slide 1.6.2 We are familiar with image errors in our day-to-day life. When we take photographs with hand-held cameras, we often encounter distortions because the subject or camera has moved, or because sunlight glints in from the side, or because the contrast and brightness of the image are poor compared with reality.

Exactly the same is true with the images recorded by satellite sensors. Before we can make sensible use of those images, we need to understand the sources of error and distortion and the means by which we can correct them to create an image which is as close to the actual landscape as possible.

Slide 1.6.3 In this lecture we will concentrate on the most important types of distortion that frequently occur in remote sensing imagery. Often, they are corrected before the images are sold to the user and, in your work, you may not be particularly aware of them. Nevertheless, it is important that we understand how distortions arise and the means by which they corrected. In some cases, most notably when images are recorded using aircraft and drone platforms, you may have no alternative than to correct the errors yourself.

An important reason for understanding the corrections of distortions in image geometry is that the techniques we employ for correction can also be used to register images to maps, and images to other images.

As with any computer-processing in image analysis and remote sensing, if you understand the basis of the algorithms employed you are in a much better position to understand their limitations and characteristics.

Slide 1.6.4 Distortions can occur in the geometry of an image and in its brightness. We will look at each of those in turn, commencing with brightness distortions.

Distortions in brightness are generally known as radiometric distortions. Radiometry, you will recall, describes the brightness levels in an image.

As seen in this slide we subdivide radiometric distortion into three sub-categories: those caused by the properties of the instruments themselves; those caused by the shape of the solar the radiation curve—i.e. the Planck radiation graph for solar radiation; and those caused by the effect of the atmosphere and particularly how the effect of the atmosphere varies with wavelength.

Any radiometric characteristics that are wavelength dependent will cause relative brightness distortions among the bands. That is a critical concern because band-to-band relative brightness differences are what allow us to understand the spectral reflectance characteristics of earth surface types. Recall the differences between the spectral reflectance curves for vegetation, soil and water from the previous lecture. We don't want those relativities upset by radiometric distortions.

Slide 1.6.5 Let's start by looking at the effect of instrumentation errors on brightness. We are particularly interested in the behaviour of the radiation detectors that measure the upwelling energy in each band.

As seen on the bottom left hand side of this slide the ideal radiation detector has a linear characteristic that indicates the magnitude of the electrical signal generated by a certain level of incident radiation. Ideally, the curve has to be linear so that variations of the output signal mirror those of the input radiation level.

There are two important detector properties described by this graph. The first is its slope, which is referred to as the ***gain*** of the detector. A higher gain detector will generate a larger electrical signal for a given input level of radiation. The second property is the ***offset***, which indicates that the detector will generate a very small signal even if there is no radiation input. That is just a characteristic of electrical devices and sometimes called *dark current*. It can be reduced by cooling the detector.

If a particular imaging sensor consists of a number of detectors (such as the ETM+ and the HRV family) it is important that their gains and offsets be matched as closely as possible. In practice, however, the situation is more like that shown on the bottom right hand side of this slide. Although the drawing is exaggerated, the detectors will often have slight variations in their offsets and slight differences in their gains.

In the simple case of the Landsat multispectral scanner there are six detectors per band. If there are noticeable differences in gains and offsets it is possible to see striping in an image which repeats every six lines.

Slide 1.6.6 This side gives an example of the line striping in a Landsat multispectral scanner image (at the top). It is very noticeable in the water part of the image.

How can we reduce the line striping distortion caused by the mismatch in sensor characteristics?

A simple method is to compute the histograms of the signals recorded by each individual detector and then match the means and standard deviations of those histograms. We will not go into detail here, but it is a fairly simple process and is often employed in practice. When we get to image contrast modification in a later lecture you will be able to see how this is done.

When that procedure is applied the "corrected" image on the bottom of the slide results. You will see that the striping has been reduced considerably.

Slide 1.6.7 Next, we examine the variation with wavelength of the solar radiation curve.

As seen here, over the optical range of wavelengths there can be a significant change in the level of radiation falling on the Earth's surface. That would suggest that the visible wavelengths would appear brighter than those in the infrared just because of the level of sunlight and not because of the spectral reflectance variation of a given the cover type. Clearly, that solar variation needs to be corrected if meaningful spectral reflectance information is to be obtained in a recorded image.

Correction can be carried out by using the ideal Planck curve for broadband sensors. For hyperspectral instruments, in which the spectral resolution is very high, the actual solar curve measured above the earth's atmosphere needs to be used because the real solar spectrum contains fine spectral detail (Fraunhofer lines).

Slide 1.6.8 Now let's look at the effect of the atmosphere on the brightness of an image.

The atmosphere absorbs radiation on its path from the sun to the pixel and from the pixel to the sensor, as we saw previously.

The atmosphere also scatters radiation directly into the sensor as depicted in this slide. There is a third, more subtle, effect and that is scattering from the atmosphere onto the pixel via a diffuse and not a direct path, also as seen here. There are other less obvious scattering pathways as well, all of which need to be considered when looking at the effect of the atmosphere on recorded image data.

Slide 1.6.9 To make the situation simpler consider just the two mechanisms shown here. We know that the direct path from the sun to the pixel and to the sensor suffers selective absorption. But what does the atmospheric scattering pathway do to the recorded signal?

Slide 1.6.10 The answer to that question is quite complex and depends on the size of the particles that make up the particular atmosphere through which imaging takes place.

Here we show three different atmospheres. At the bottom we have a clear atmosphere in which the only scattering elements are the molecules of the gases that make up the atmosphere itself.

In the middle picture the atmosphere also contains some smaller particulates such as mist.

At the top the atmosphere is foggy, containing large scattering particles of water.

Scattering from a clear atmosphere is called Rayleigh scattering. It is strongly wavelength dependent as illustrated in the graph, with blue light being scattered the most. The net effect is a blue colouring; that is why we see the sky as blue on a clear day.

Scattering from a light haze is called Mie scattering. Is not as strongly dependent on wavelength as Rayleigh scattering, so that the effect is a bluish-white appearance.

Scattering by large particulates is almost independent of wavelength so that the atmosphere then appears almost white. That is why we see fog as white.

These scattered signals superimpose on the direct path signal from the previous slide. As you might expect, the short wavelengths are affected more than the long wavelengths.

Slide 1.6.11 One simple means for compensating the wavelength-dependent effect of atmospheric scattering is to produce histograms of each of the recorded Image bands.

The effect of atmospheric scattering shows up in the histograms as offsets from zero, as illustrated in the histograms in this example.

Recall that the shorter wavelengths are affected more. This is seen in the visible blue histogram in this example. As wavelength increases the offset due to the atmospheric scattering reduces, as depicted.

In order to correct for the effect of the atmosphere the offsets are subtracted in each case so that the image histograms are reset to start from zero.

Slide 1.6.12 Correcting for the effect of atmospheric absorption on the direct path signal usually requires a knowledge of the wavelength dependence of the various actual absorbers in the atmosphere. There are a number of quite comprehensive software packages that can be used for that task.

Slide 1.6.13 This slide is an illustration of the effect of all atmospheric constituents on the spectrum of a vegetation pixel at the top, which also includes the distortion caused by the shape of the solar irradiation curve. The corrected version is at the bottom.

Note that complete absorption due to water vapour cannot be compensated, but that is not a problem because there is usually no significant information available in any case in those regions.

Slide 1.6.14 As we noted at the start of this lecture many of the distortions discussed will have been compensated in the image product you purchase, particularly from satellite missions.

Nevertheless, as a remote sensing practitioner, it is essential that you have some background knowledge of what constitutes errors in radiometry so that you are mindful of what corrections have been carried out and what might be needed in the case of imagery you acquire which has had no corrections applied to it.

Slide 1.6.15 In the first of these questions use your own experience, particularly if you have flown over regions of patchy fog and mist.

The last question, in a sense, asks you to build up what might actually be recorded by a sensor as against what we think might be recorded.

Lecture 7. Geometric distortions in recorded images

Slide 1.7.1 We now come to the rather more important matter of geometric distortions in remote sensing imagery. This is important, because images from aircraft and drones are often not corrected for these sorts of error, so you need to know how to do that yourself, if geometric integrity is important.

Slide 1.7.2 As we mentioned earlier the techniques used for correcting errors in geometry are the same as the methods we use for making sure an image is registered to a known map basis.

There are many sources of geometric error including the fact that the earth and platform are in motion during image acquisition; that the platform can vary in altitude, attitude and velocity; the fact that the earth is curved when seen from space; and the fact that the instruments themselves introduce geometric variations by virtue of their design.

We make the assumption that all recorded bands are affected equally. Therefore, we apply our techniques to each band in an image, separately, but we only have to develop our correction methods on a single band.

Slide 1.7.3 It helps for much of what is to follow if we look carefully at how an image is built up from recorded pixels. The diagram in this slide shows a set of pixel or grid centres. The grid centres are spaced apart by the IFOV of the sensor and, in this case, we have shown an array of L lines of M pixels each.

What is a pixel? It is a sample of the earth's surface equal in size to the IFOV of the sensor. A set of pixels is a set of those samples.

The rate of sampling is arranged, ideally, so that when we form an image by laying down our recorded pixels on the grid the pixels join up with each other as depicted here.

We need to keep this model in mind in everything we now do with respect to image geometry.

Slide 1.7.4 The first time this rectangular grid concept becomes useful is in understanding the error introduced into an image as result of the rotation of the earth.

The signal coming from a remote sensing platform is just a string of pixels for each recorded line of data. Knowing nothing better we just lay them down on our regular image grid, as discussed in the previous slide. That gives the square-looking image on the left-hand side of this slide.

However, during satellite image acquisition, the earth is moving to the east underneath the platform. In this case imagine the platform is moving down the slide.

The part of the image that is recorded last was actually to the west when the recording of this image segment started, and has moved itself directly under the platform by the time the last line of pixels is recorded. Therefore, to make the image geometry reflect that of the actual portion of the earth's surface imaged, the second and later lines of pixels need to be shifted progressively to the west, as shown in the right-hand image.

If we know the velocity of the spacecraft, the earth velocity at that location and the size of the image it is relatively easy to work out how much each line of pixels needs to be shifted by.

Slide 1.7.5 We now consider a very unusual form of geometric distortion. It occurs when the scan of the sensor is very wide. In other words, it has a large FOV.

The diagram on the left-hand side shows the relevant geometry. Notice that the area of earth imaged at the swath extremity is much larger than that at nadir—that is, directly under the sensor. And yet, even though it corresponds to a larger region on the ground, the actual pixel is displayed as the same size as the pixel at nadir when used to construct an image.

When those recorded pixels are placed on the regular display grid, the pixels towards the swath edges, even though the same displayed size as those towards the centre of the scan, cover a much larger ground region, and thus appear compressed by comparison to those at the swath centre.

To illustrate what this does to an image consider the ground scene on the right-hand side of this slide. Be careful here. The rectangular grid shown might represent a set of equally-sized square fields. It is not the display grid. Imagine that each of those fields consists of many pixels. The two broad diagonal lines might represent roads that cut across the fields at 45 degrees, as shown.

When the recorded pixels are laid down on the regular display grid, the effect is to compress the edges of the image as shown on the very right-hand side image. That is because the pixels at the edges represent a greater geographical region than those at the centre, and yet are displayed as the same size. The visual affect is to give the impression that the image rolls off at the edges. The diagonal roads take on the S shape shown. Sometimes this panoramic distortion is called S-bend distortion.

Slide 1.7.6 From satellite altitudes, sensors with a wide FOV are sensitive to earth curvature, as seen in this slide.

Without going into too much detail the net effect on the image is to exaggerate the panoramic distortion of the previous slide.

Slide 1.7.7 Now let's look at some of the errors caused by the instruments themselves.

For an oscillating mirror scanner, as seen here, the mirror has to slowdown, stop and reverse at the end of each scan. By definition, the scan has to be nonlinear at the swath edges. To minimise the nonlinearity a scanning window is defined over which data is recorded and beyond which any other detected radiation is ignored. Clearly the scanning window is chosen to exclude the extreme non-linearities at the scan edges.

Slide 1.7.8 We now look at two other instrumentation problems.

The first is related to the fact that some scanners take a finite time to cross the swath, while the platform is moving forward during data acquisition. That means that the portion of the earth being scanned is further forward on the swath edge at the end of the scan. To correct for this effect the display pixels have to be shifted progressively backwards towards the edge of the image.

The second error is called aspect ratio distortion. It is caused by the pixel sampling rate across the scan line not being synchronous with respect the IFOV of the instrument. If the sampling rate is too fast the pixels overlap. If it is too slow the pixels have spaces between them. Again, when these pixel sequences are displayed on a regular grid the effect is to expand or compress the image across the track.

Slide 1.7.9 The last point in this summary demonstrates the expansion of the image across the track caused by oversampling with respect to the IFOV of the instrument. This is a particular problem with the Landsat multispectral scanner.

Slide 1.7.10 When answering the questions in this quiz you should keep in mind that the displayed version of the image almost mirrors what is happening on the ground in terms of geometric errors. For example, to correct for the effect of earth rotation eastwards the pixels have to be shifted to the west.

Lecture 8. Correcting geometric distortion

Slide 1.8.1 We now come to methods for correcting errors in the geometry of a recorded image. This is a fairly large topic and will occupy the next two lectures – using two different approaches.

Slide 1.8.2 Both methods for compensating or correcting geometric errors in remote sensing imagery are used in practice.

The first involves setting up mathematical equations which describe the types of distortion and then using those equations to correct the distortions. That can be a complicated process if many different sources of distortion are of concern, but it is regularly used for a systematic correction by satellite operators. It also depends on a good knowledge of the ephemeris—that is the velocity, position and pointing with time— of the platform.

The second method accounts for all sources of distortion in a single operation. It avoids the need to model mathematically the various types of distortion and instead uses a mathematical relationship between where pixels appear in an image and where they should appear in a map. This is a very common approach and has the added advantage that the map basis can be chosen according to the projection preferred by the user.

Slide 1.8.3 We will illustrate the first approach by using the error introduced into an image by the rotation of the earth during image acquisition. This slide shows the image laid down on a rectangular grid, which is known to be incorrect geometrically, and beside it the version which has been compensated for this rotation.

To start the process we need to define two coordinate systems—one for the image as recorded, and one for the corrected version.

Slide 1.8.4 The coordinate systems have their origins at the top because the platform is assumed to be moving down the page as it acquires its image. This is equivalent to a satellite in a descending node.

We use the coordinates u and v to describe the recorded image laid on the regular grid, and coordinates x and y to describe the pixels in their correct positions with respect to the earth's surface (i.e. the chosen map grid).

Clearly, what we have to do now is to form a mathematical relationship between the pairs of coordinates.

Slide 1.8.5 We now set up two equations. They describe the position of a pixel in the correct coordinate system as a function of the position of the pixel as recorded.

Note that the line coordinate (v and y, respectively) does not change. Therefore, we get the second (unchanged) equation shown at the bottom of this slide.

However, to compensate for earth rotation the coordinate across a scan line has to be adjusted progressively to the west as we increment down the lines of the image. In other words, as well as being a function of u, x must also be a function of v, representing this shift.

The coefficient alpha describes the degree of correction to the west as the platform moves southwards.

Slide 1.8.6 It is convenient to express the pair of equations in matrix form, because that helps when we have to make several corrections.

The value of alpha depends on the platform orbital properties and the location at which the image was recorded.

Slide 1.8.7 If we examine the geometric characteristics of other distortion types it is possible to set up similar equations, as shown on this slide.

The next step would be to use these relationships to create a geometrically correct image. That requires inverting them to write u and v in terms of x and y. Because the technique we are going to treat in the next lecture is so flexible in terms of correcting any types of geometric distortion, we will not proceed any further with mathematical modelling.

Slide 1.8.8 We have only reviewed the most significant sources of geometric distortion here. Please consult some standard textbooks to see the very large range of mechanisms that can cause geometric errors in an image.

Slide 1.8.9 The first question here has been treated in the lecture. The idea of the question is for you to derive the matrix form of the correction.

The third question is particularly interesting. Later in this course we're going to treat methods for attaching labels to pixels indicating what they represent in terms of ground cover types. This question asks you to consider whether the geometry of an image should be corrected before that labelling process is carried out, or whether the labelling should be done first and then the labelled pixels corrected for their geometric positions.

Lecture 9. Correcting geometric distortion using mapping functions and control points

Slide 1.9.1 This lecture introduces a very powerful and common method for correcting errors in geometry. As noted earlier the technique also allows us to register one image to another image or an image to a map.

Slide 1.9.2 It is an alternative technique to the mathematical modelling approach of the previous lecture. It is based upon two principal requirements.

First, we assume we have available a map of the region which is correct geometrically and is in scale not too different from that of the image we wish to correct.

Secondly, we need to be able to identify points—called control points—in both the map and the image so that we can tie the two together mathematically. The real power of this technique lies in setting up mathematical equations that link points in the image with corresponding points in the map.

We will use the control points to set up those mathematical equations. In a sense we are assuming that if we can relate the control points from the image and the map mathematically, and that the control points are indicative of the map and image geometries, then other points in the map and image will also be related by those same equations.

Slide 1.9.3 This slide shows what we have in mind. The geometrically correct map is on the left-hand side and image to be corrected is on the right-hand side. Notice that these are in the opposite positions from the treatment of the previous lecture. It is just more convenient in this analysis to have them in this new order.

As before, we have chosen the coordinates x and y to describe the position of a point or pixel in the map, and the coordinates u and v to describe position in the image.

In principle, it should be possible to set up equations that relate the coordinates of a point in the image knowing its position in the map. This is shown here by the pair of functions in which **both** of the image coordinates are shown as functions of **both** map coordinates.

Those functions will contain unknown coefficients which, when given explicit values, will allow us to go between the map and the image. To find those explicit values we need some points in common—i.e. simultaneous sets of x, y and u, v—that we can substitute into the equations. Those points are called control points, or sometimes ground control points (GCPs).

Slide 1.9.4 Here we show the relationship between points in the map and the corresponding points in the image by drawing a grid of pixel centres in each case. What we are going to do is choose a grid position on the map and, by using the mapping functions, find the corresponding grid position in the image.

What sorts of functions should we use for this? The simplest are polynomials.

Slide 1.9.5 Here we show the mapping functions expressed as polynomials, in this case of order or degree two. In principle, any order could be used, expecting that higher orders will give more accurate results. However, high order polynomials introduce their own problems as we will see later.

The unknown coefficients to be found using the control points are the a_0, b_0 etc.

Slide 1.9.6 If we assume that we have used the control points and identified the unknown coefficients, then we use the two mapping polynomial is in the following manner.

We move over the map grid point by point (i.e. grid intersection by grid intersection), and for each point use the mapping polynomial is to find the corresponding point in the image. We find the pixel value at that point and transfer it onto the map position we started with.

That is done for all grid positions on the map thereby building up a geometrically correct image, essentially using the recorded image as a source of pixels.

Slide 1.9.7 This slide shows the process broken into the two individual steps. At the top we find pixel locations in the image which correspond to pixel positions on the map. On the bottom we transfer the pixel from the image and onto the map grid.

This is just a small point, but it is worth noting: the pixels that are centred on an image grid (map or recorded image) are of such a size that adjacent pixels join up against each other.

Slide 1.9.8 There are, now, two important matters to be considered before the method is workable. The first is how to determine the unknown coefficients in the mapping polynomials. As said earlier, that involves the use of control points.

The second practical matter is that the mapping from a map grid position using the polynomials may not project exactly onto a pixel position in the image grid. That will form the focus of the next lecture.

Slide 1.9.9 Remember that the control points are pairs of coordinates in the map and the image. Essentially, each control point is represented by the set of values of x, y, u and v. If we had, say, 20 control points, we would have 20 sets of those four numbers.

What we do is substitute those known values into the mapping polynomials and thus set up a set of simultaneous equations which when solved yield the required values of the unknown coefficients.

Slide 1.9.10 Clearly, we need to have at least as many control points as there are unknown coefficients in one of the mapping polynomials. We can use the same set of control points to generate the unknown coefficients in the other mapping polynomial – we do not need a separate set.

For a second order mapping polynomial a minimum of six control points is required. Three are needed for a first-order (linear) mapping polynomial and 10 are required if third order mapping polynomials are used.

Usually, we choose more control points than necessary and generate least squares estimates of the unknown coefficients.

A particular advantage in choosing greater than the minimum number of control points is that any specific control point which may not be well-placed in either the map or the image will not have an undue influence on the values of the coefficients generated.

Slide 1.9.11 Effectively, this technique depends upon having an accurate way of relating points in a map to the corresponding points in an image. We do this by the process of choosing control points and establishing mapping polynomials. We will see an example in a later lecture.

Slide 1.9.12 Pay particular attention to the second question. You might like to think about the similar problem of fitting a curve through a set of data points, using curves of increasing order. Will a straight line do a better job than a cubic polynomial, or not?

Lecture 10. Resampling

Slide 1.10.1 We now introduce a new term in this lecture. The process of finding an actual value for the pixel in the image to place on to the map grid using the process of mapping polynomials is called resampling.

Slide 1.10.2 Remember we determined the explicit mapping polynomials using the set of control points to find the unknown coefficients. We then use those polynomials to find points in the image which correspond to points in the map. If the point in the image indicated by the mapping polynomials falls exactly on a pixel grid centre then the pixel is simply relocated on to the corresponding or correct position in the map grid.

However, rarely will that be the case. More often the point in the image indicated by the mapping polynomials will not fall exactly on a pixel centre in the image but somewhere between several pixel centres, as we see in the next slide.

Slide 1.10.3 Here we show the mapping from the geometrically correct map to the image for the general case where the projected image coordinate does not match a pixel grid centre exactly. Instead, the projected point lies somewhere in a region bounded by four grid centres.

What we have to do, therefore, is to estimate what the brightness of the original scene would have been at that position. Effectively, that means carrying out an interpolation over the neighbouring, measured pixels.

Note here we have indicated the pixels explicitly by dashed lines centred on each of the grid intersections.

Slide 1.10.4 How do we go about finding a brightness for a pixel (i.e. for the scene) at the location indicated by the mapping polynomials?

A simple approach, and one which is often used in practice, is to choose the nearest recorded pixel.

This is called nearest neighbour resampling. For it to work effectively, the grid spacings in both the map and the image should be about the same.

Slide 1.10.5 There are two other resampling techniques often used in practice. Both are based on interpolations of brightness over neighbouring pixels.

The first involves linear interpolations over the four surrounding pixels. It starts by computing the two interpolated brightness values along the scan lines above and below the projected pixel point, using the real pixel values either side. These are shown as the "intermediate interpolants" in the top right-hand diagram. The next step is to do a vertical interpolation over those two intermediate values to get a final brightness value for the pixel. This process is called bilinear interpolation.

The second, more sophisticated, procedure is to interpolate over four sets of four pixel brightnesses as illustrated on the bottom right hand side of this slide, and then over the four vertical intermediate interpolants. Cubic polynomials are used to undertake the interpolation so that the technique is generally called cubic convolution resampling.

While a more complicated process, cubic convolution resampling generally leads to a smoother looking image after geometric correction. By contrast, nearest neighbour resampling can produce a blocky looking product if the scales of the image and map are significantly different.

Slide 1.10.6 Although we have developed the process of using mapping polynomials for registering an image to a map, and therefore effectively removing the major sources of geometric distortion, the same process can be used to co-register sets of images.

This is often needed if the user has available for analysis several images of the same geographical region, often taken by different platforms and sensor types.

As noted here we're not restricted in how many images we can co-register. It is just a matter of choosing one image as a master (taking the place of the map) and registering all other images to it.

Slide 1.10.7 As seen in this summary the process of registering an image to a map in order to remove geometric errors is often called geocoding or sometimes georeferencing, because the pixels can now be represented by their map coordinates—latitudes and longitude—rather than by the original row and column numbers.

Slide 1.10.8 The questions here ask you to think about some practical aspects of using ground control points and mapping polynomials for geometric correction. Please give them serious thought because they will feature strongly in the next lecture.

Lecture 11. An image registration example

Slide 1.11.1 In this lecture we will demonstrate the application of the procedure just developed by registering one image to another. The same procedure is used for registering an image to a map to remove geometric distortions. By using an image to image example, it is easy to highlight some pitfalls that must be avoided.

Slide 1.11.2 The example we have chosen involves portions of two Landsat multispectral scanner images of a region in the northern suburbs of the city of Sydney, Australia. One was acquired in 1979 and the other in 1980.

We have only shown one of the near infrared bands here. Once they have been registered all bands can be registered using the same control points and mapping polynomials to generate registered colour products.

Clearly there are major brightness differences between the two images. Since they have been recorded in the near infrared both healthy green vegetation and urban regions show as bright. Water shows as black. And the dark grey regions are burnt vegetation resulting from fires.

In the 1979 image there is a major fire scar in the centre. By 1980 much of that has revegetated but two new fire scars are evident, one to the northwest on the boundary of the river, and the other in the central southern part of the scene.

There are geometric differences between the images as well, even though they are less evident. It is by registering the 1980 image to the 1979 image that we will remove the geometric differences between them.

Had the 1979 image been a map then that would amount to removing geometric distortion from the 1980 image.

Slide 1.11.3 Our first task is to choose a set of ground control points. On this slide we have shown two different sets. Those on the top pair of images are well distributed over the scene, whereas those in the bottom images tend to be clumped, with none towards the image edges.

Notice that we have chosen some control points on the water-land boundaries. As much as possible, those sorts of control points should be on headlands and not river inlets so that tidal variations are minimised.

In general, we should choose as control points features which do not change in position from image to image. Road intersections are good as are corners of buildings and fields.

In this example we used third-order mapping polynomials and cubic convolution resampling.

Slide 1.11.4 This slide shows the results of registering the 1980 image (slave) to the 1979 image (master) using the good set of control points—that is the set that was well distributed over the image.

In order to demonstrate how well the registration has worked we have displayed the master image as red and the slave image as green. The reason for doing that is that where the images are in alignment geometrically and have approximately the same brightness, the combination will appear yellow, which is the additive combination of red and green.

Here we see that the registration is very good. Apart from the colour differences due to the 1979 and 1980 fire scars, the combination is yellow, with no obvious red-green separation. The outer red border is just a part of the 1979 scene for which there is no 1980 coverage.

The fire scar colours are interesting. The 1979 scar shows as green because there is very little red signal from the 1979 pixels, whereas the 1980 scars show red for the same reason—i.e. there is very little 1980 signal in those portions of the image.

Slide 1.11.5 This slide shows the result of the registration using the poorer set of control points. Where the control points were bunched the registration is very good. But beyond the distribution of control points the registration is particularly bad. Note especially the red and green offsets around the edges of the water.

Noting the yellow colour, we can see that the 1980 image has been stretched out of shape across the diagonal.

We now wish to understand that behaviour.

Slide 1.11.6 We can get a good idea of well-behaved and poorly behaved registration by looking at fitting a curve through a set of points, as shown in this slide.

The first order equation matches the distribution of points moderately well and seems to extrapolate in a well-behaved manner beyond the set of points. The second-order curve gives a slightly better match to the points but seems to diverge unnecessarily beyond the data points. Finally, the third-order polynomial can be made to fit the data points more closely than the other two curves, but it behaves quite erratically beyond the extent of the data—in other words, it does not extrapolate well.

The general lesson we learn from this exercise in curve fitting is that low order polynomials may not fit a set of data points as well as a higher order polynomial but in general terms they will extrapolate better. In contrast, higher order polynomials may match the available data better but will extrapolate poorly.

Returning to the previous slide the third order mapping polynomial fits the region of the image in the cluster of control points very well but beyond that region it extrapolates poorly.

That gives us a particularly important guide in the choice of control points. As far as possible, the set of control points should be distributed uniformly across the scene and enclose the region of the image for which accurate registration is important.

If there is any doubt about the distribution of control points a lower order mapping polynomial should be used.

Slide 1.11.7 This example demonstrates the benefit of using lower order of mapping polynomials when the distribution of control points is not good.

It uses the poor set of control points from the previous example. The result of that example is shown on the left-hand side. Remember, this was generated using third-order mapping polynomials.

Without changing anything else, the image on the right-hand side was generated using first-order mapping polynomials. As noted, the extrapolation of the registered 1980 image (in green) is much better than when the third order polynomials were used.

There will be greater registration error in the vicinity of the control points, but that is not noticeable visually in this simple case.

Slide 1.11.8 Just by way of a reminder, every band of an image has to be registered or corrected in this manner. Provided the band-to-band registration from the image supplier is good, the same control points and mapping polynomials can be used.

Slide 1.11.9 These questions relate to practical aspects of image registration, including geometric correction, and not so much to the theory of the process itself. They ask you to think about where you might generate unintentional problems if you are not careful with your choice of technique, scale and methodology.

Lecture 12. How can images be interpreted and used?

Slide 1.12.1 We are now at a very important stage in the course. In the previous lectures we have learnt about how sensors are used to record images, the nature of reflected sunlight and earth thermal emission as sources of earth data, how the atmosphere affects the transmission of radiation from the sun to the surface to the sensor, and how we can correct for image errors in brightness and geometry.

Slide 1.12.2 Effectively, that means we are now at the stage of having acquired good quality data ready for analysis. The rest of our course depends upon having reached this stage satisfactorily.

Slide 1.12.3 There are two very broad approaches that can be taken when endeavouring to interpret images recorded by a remote sensing platform.

The first involves a person doing most of the work. A person skilled at interpreting an image visually is called a photointerpreter and the process is called photointerpretation. This is a long-standing method for interpreting images that goes back to the days of air photo interpretation. The analyst builds up substantial knowledge of how to look at the various cues and clues in an image in order to extract meaningful information.

The second approach, which is much more recent, involves the use of sophisticated computer algorithms to help undertake an analysis. A human is still involved but their skills are more in knowing how to get the best results out of the computer-based approach rather than being able to interpret the images of themselves.

When computers are involved in the interpretation process, the approach goes by a number of names, including quantitative analysis, classification or machine learning. More recently, artificial intelligence methods have been contemplated for remote sensing image analysis. We will say something about that later in our course.

We will now examine both photointerpretation and qualitative analysis in general terms but the major emphasis of the remainder of this course is on the computer approach.

Slide 1.12.4 Let's start with photointerpretation. A skilled analyst works with a colour image product composed from the recorded image bands, or sometimes with the individual bands themselves displayed as black and white.

Irrespective of whether black and white or colour imagery is used, the analyst makes use of colour or relative differences in brightness from band to band, and relationships between objects and pixels. Changes with time of features from one image to another in a co-registered set are also important clues for the photointerpreter.

Slide 1.12.5 Because an analyst relies on understanding brightness and colour, it is important that we know how colour image products are formed if we want to appreciate the nature of the products with which a photointerpreter works.

As you probably appreciate from your own knowledge of digital photography or colour television, a colour image is made up of three black and white images displayed on each of the three primary additive colours of blue, green and red. Keep in mind that an individual band of image data is always represented from black to white (lowest brightness to highest brightness), even though it might represent, say, the reflectance of the earth's surface in the red wavelength range (seen through a red filter, for example).

A challenge in remote sensing is to know which bands to display in which colours. We only have three display colours but we could have tens or hundreds of bands of image data from which to choose.

Slide 1.12.6 We illustrate in this side three different colour images formed from different band combinations, using data recorded by the HyMap hyperspectral scanner.

Remember, you have to choose which bands to display in each of the red, green and blue colour primaries.

On the left-hand column of images we have chosen a recorded visible red band to display as red, a visible green band to display as green, and a visible blue band to display as blue. The result is a natural colour image where the vegetation appears green and urban and other developed surfaces take on a brownish or slightly blueish appearance.

In the centre column we have created what is commonly called a colour infrared image. It is based upon the choice of bands long-used in colour infrared aerial photography. A near infrared band is displayed as red, a visible red band is displayed as green and a visible green band is displayed as blue. Although this sounds unusual, it gives a very rich colour product in which healthy vegetation (having a high near infrared response) shows as red and bare regions show as shades of blue and green. Clear water generally displays as black.

In the right hand column we show a colour image product made from a set of infrared bands. As seen in the colour image there is a fairly good differentiation of different cover types.

Often an analyst will experiment with bands to display, guided by their knowledge of the spectral reflectance characteristics of the earth's surface cover types of interest in a particular interpretation exercise.

Are there any conventions observed when forming colour image products?

The only real convention is that the bands are displayed in the same order of wavelength as the primary display colours. Therefore, the recorded band with the longest wavelength is shown as red and that with the shortest wavelength is shown as blue.

Slide 1.12.7 Once we form the colour Image product we have to work out how to interpret it. That requires reference back to the spectral reflectance characteristics of the different earth surface cover types present in an image.

On this slide we have shown where three bands (green, red and near infrared) are located on the reflectance spectra. When combined into the colour infrared composite, vegetation appears as red, soil and urban regions appear as blue and green, and water will appear as bluish black depending on its depth and turbidity.

There is however an intervening step which you may have picked up by a careful examination of the spectral reflectance curves and the colour infrared product. From the spectral curves it looks like all cover types should be predominantly reddish. Indeed, if the composite were formed from the raw data recorded by the platform that will be the case. In practice, however, we undertake what is called a contrast enhancement step in order to get the colour product shown on the bottom right hand side of this slide. We will see how to do that in the next lecture.

Slide 1.12.8 In the early days of remote sensing, when instruments had only three or four wave bands, selection of which three to use when forming a colour composite product was straightforward. With modern instrumentation, however, the analyst is faced with a more complicated task to choose the bands to display. In some ways, this places more dependence on the analyst's knowledge of the spectral characteristics of the scene properties of interest.

Slide 1.12.9 These quiz questions are important in terms of drawing your attention to the limitations of a human analyst, limitations which we expect will not be a problem when using a machine to assist with the interpretation.

Lecture 13. Enhancing image contrast

Slide 1.13.1 This lecture is a little outside the mainstream of our treatment, but it is important because most of the images we deal with visually have had their contrasts enhanced in order to improve their visual quality and range of colours. Here, we want to understand a little of how that is done.

Slide 1.13.2 The detectors used to record each band of image data are calibrated so that they can respond to the darkest and brightest cover types anticipated. Typically, those cover types could be deep clear water or, snow and cloud tops.

That means that most of the cover types that are imaged fall in a range of brightness well inside the range from total black to full white as shown in this side.

Slide 1.13.3 For a typical recorded image, the appearance is therefore slightly dull and poor in contrast. It would be good if we could expand the range of recorded brightness so that the signal of interest covers the full range from back to white.

What we need to do now is understand how that can be made to occur.

Slide 1.13.4 In order to set up a process for changing the range of brightness values we need, first, some means for describing the distribution of brightness in an image (i.e. a band). We choose for that the image histogram which is a bar graph of the number of pixels of a given brightness plotted as a function of brightness value.

Slide 1.13.5 Here we see an image as it has been recorded by an aircraft scanner. Note how poor it is in contrast, brightness and colour. Note particularly that it has a reddish appearance as we mentioned in the last lecture.

On the right-hand side, we have shown three histograms, one corresponding to each band, and coloured according to the colour in which the band is displayed in the colour composite image.

Note that while each histogram commences close to the lowest brightness in each case—i.e. zero–the upper extremities of the histograms are well short of the maximum available brightness value of 255.

Ideally, we would like the histograms to cover the full range of brightness.

Slide 1.13.6 If we could stretch out the histogram, as shown here, so that it covers the full brightness value range, we would have an image with much better contrast as we will see shortly.

Slide 1.13.7 To do that we introduce a graph, called a brightness value mapping function. It shows us how to take the old rightness values in a histogram and change them to new values which cover the full brightness value range. Effectively what it does is tell us how to relocate the bars in the histogram more favourably.

We usually work out the nature of the brightness value mapping function by examining the recorded histogram and noting the desired new histogram distribution. Once we have that mapping function, we apply it to the image in the following manner.

We start with the first pixel in the image read its value onto the abscissa of the mapping function and read off its new value from the ordinate. That new value is used in place of the old brightness value of the pixel. That is done pixel by pixel and line by line until all of the pixels in the image have had their brightest values modified according to the brightness value mapping function. The resulting image then makes a much better use of the available range of brightness.

We have to do that for each band of data. The brightest value mapping function will be different in each case.

Slide 1.13.8 Here we see the results. Commencing with the poorly contrasting original image we construct the histogram for each band, then expand those histograms via their own respective brightness value mapping functions. When applied to the pixels in each band the colour composite image so generated is the much nicer looking image product on the right-hand side.

In the final product we see a much better range of brightness and contrast and certainly a much better use of colour.

You can see now why contrast enhancement is almost always used in remote sensing when visual interpretation is important.

Slide 1.13.9 Because we are dealing with a discrete number of brightness levels when handling remote sensing imagery, the brightness value mapping functions of the previous slides are actually set of points equal in number to the number of discrete brightness values.

For 8-bit imagery, there will be 256 pairs of input (old) and output (new) brightness values. For 10-bit data there will be 1024 pairs, and so on.

As a result, we represent the brightness value mapping function in a computer algorithm as a look-up table, as shown in the last point on this slide.

Slide 1.13.10 Take particular note of the second question here. The brightness value mapping function does not always need to be linear as discussed in this lecture. Instead, it can have a range of non-linear forms which can give added emphasis to particular ranges of brightness.

Lecture 14. An introduction to classification (quantitative analysis)

Slide 1.14.1 We are now at the stage in our course where we can start seriously looking at how to interpret images recorded in remote sensing missions, particularly using computer-based algorithms.

Ultimately, this will take us into a mathematical journey, sometimes quite complex, but we will ensure that if you do not have the required mathematical background the accompanying description should be sufficient for you to understand how the techniques operate.

We start, though, with a descriptive outline of the essential aspects of computer image analysis, which is often called classification, or quantitative analysis, in remote sensing.

Slide 1.14.2 The first concept to keep in mind is that we almost always work at the level of the individual pixel. There are procedures that take a different approach but the great many that we will encounter focus on the individual pixel because it is the smallest quantifiable element in an image.

Remember, for each pixel we have a set of recordings, composed of the brightness values detected in each of the individual wave bands. That set of measurements is collected together and written in the form of a column surrounded by square brackets, which we call a column vector or pixel vector.

Slide 1.14.3 In classification we use the properties of the pixel vector to help find a label for the pixel, a label that represents one of the classes of ground cover in which we might be interested.

For example, if we wanted to use remote sensing to map the natural landscape, we might employ computer classification to analyse each pixel in the image and label them as grass, soil, water, forest, etc. Once all pixels are labelled, we then have a map of cover types, which we call a "thematic Map" because it is a map of themes or classes.

Those labels are specified by the end-user. What the image analyst has to do is come up with a computer-based approach that allows those labels to be applied to the pixels through an analysis of the pixel vectors.

Slide 1.14.4 In this slide we develop several of the most important concepts in image analysis. To do so, we have assumed, for convenience, that the sensor we are dealing with has only two wave bands–one in the visible red, and one in the near infrared.

Again, for convenience, we assume that the scene being imaged by the remote sensing instrument consists of just three natural cover types: vegetation, soil and water.

Invariably, the first step in any classification process is for the analyst to find sets of pixels in the recorded image for which the cover type is known. In the upper left-hand depiction of the scene, we have indicated by the shaded shapes that the analyst knows that the pixels in those regions are actually pixels of vegetation, soil and water, as indicated.

We now need to remember something from a previous lecture, and that is the shapes of the spectral reflectance curves for the three cover types of interest. They are shown on the bottom left hand side of the slide. In each case, several examples of reflectance curves are shown, indicating that in practice there is natural variability among pixels even of the same cover type.

If our sensor takes measurements in the red and near infrared, that amounts to sampling the cover type reflectance curves at those wavelengths, as seen in the figure. For the vegetation pixels the red samples will have low brightness whereas the infrared samples will have high brightness values. By contrast, the samples for soil will be moderately bright in both bands and those for water will be moderately dark in both bands.

We now introduce one of the most important concepts in classification—the representation of pixels in a geometric space defined by the sets of spectral measurements. Such a space is shown on the right-hand side of the slide in which individual pixels are plotted according their sets of brightness values.

This is just a cartesian coordinate system in which the axes are the pixel brightness values.

If we now return to the sets of pixels for which we know the correct labels, we can plot them in our coordinate system, as shown on the diagram. Because pixels from the same cover type will have similar spectral reflectance curves, when they are plotted in our coordinate system they will tend to group or cluster, as indicated.

Effectively, those pixels for which we know the labels define regions in the coordinate system as their own. There is a region in which we expect to find water pixels, a region in which we expect to find pixels of vegetation and a region in which we expect to find pixels of soil.

In other words, the pixels for which we know the labels effectively segment the coordinate system into a set of regions corresponding to the classes of interest. We could make those regions explicit by drawing lines or boundaries that separate the classes, as indicated.

We have now introduced some terminology. Besides pixel vector, we use the term “class” to describe the particular cover type on the ground, and we call the coordinate system a “spectral space.”

The process of using pixels for which the class labels are known to find regions in the spectral space corresponding to each class is called “training”. That is the first step in classification—i.e. to use known data to find corresponding regions in spectral space.

How does the analyst know the labels for some pixels? That can sometimes be a long and expensive task, depending on the complexity of the exercise. Often, field visits are required so that the class labels for groups of image pixels can be found; sometimes, particularly in simpler classification exercises, human image interpretation, or even the analysis of accompanying air photos, can be used to establish the training pixels.

Finally, this example has been based on a two-band sensor, which has led to a two-dimensional spectral space. In practice, of course, the dimensionality can be much larger. A sensor with ten wavebands will generate a ten-dimensional spectral space. While we can easily envisage a two- and even three-dimensional space, we are not equipped mentally with the capacity to go beyond that. But that is not a problem. While we will develop many of our techniques on two- and three-dimensional examples, the corresponding models and mathematics are easily extensible to any dimension. So, all we have to do is develop our ideas in, say, two dimensions and understand that mathematically any number of dimensions (i.e. wave bands) can be handled.

Slide 1.14.5 We can now use the results of the training stage of the previous slide to help us add labels to all of the other pixels in an image—that is, those we do not currently have labels for.

This slide shows the process. We take an unknown pixel and plot it in the spectral space according to its brightness values. Because, using training pixels, we previously had segmented the spectral space into regions representing each of the cover types, we can now add a label to the new pixel by seeing where it falls in the space. In this particular case we have labelled a vegetation pixel.

As we noted, the step in the previous slide was called training; this second step is called classification, labelling or generalisation.

Slide 1.14.6 In summary, classification or quantitative analysis is a two-stage process. The first is understanding the structure of the spectral space by using sets of training pixels. The second step is to use the results of training to enable us to add labels to pixels we do not yet know about.

Looking at the diagrams in this lecture we could create a simple classifier by setting up the equations which describe the lines which separate the different segments in spectral space. Indeed, a very simple and early classifier did just that by making the lines midway between the mean positions of two groups of pixels (see the quiz questions following). We’re going to look at much more complex classifiers because of their improved performance.

Slide 1.14.7 The questions in this quiz focus your attention on the structure of the spectral space. Even though most of the time the dimensionality is too high for us to envisage, it is still nevertheless important to understand, as much as possible, the structure of the data domain with which we’re dealing.

Lecture 15. Classification: some more detail

Slide 1.15.1 In this lecture we will consolidate some of the concepts from the last and compare the attributes of a human interpreter undertaking image analysis with that analysis being carried out by a classification algorithm.

Slide 1.15.2 Remember, we are dealing with a two-stage process for machine labelling–training and classification. In the first, a human interpreter is involved in identifying training data (as well as selecting the particular classifier to use—see later).

Let's now look at what a human can do in image interpretation that a machine currently finds difficult, and vice versa.

Slide 1.15.3 In this slide we compare the performance of a human interpreter to that of a computer-based algorithm from a number of important perspectives.

First, when an image can consist of thousands to millions of pixels it is impractical for a human to try to work, in general, at the scale of an individual pixel. By comparison, because of the speed of a computer, working at the pixel level is practical and normal.

Because we estimate area in an image by counting pixels, if a human interpreter cannot easily work at the pixel level then the ability to estimate areas is limited. Again, by contrast, area estimation by computer is very easy.

We have already discussed this next one to some extent but when a sensor has more than three bands it is difficult for a human interpreter to maintain attention over all bands when carrying out an analysis. A machine, by comparison, can handle any dimensionality provided we set up the mathematics suitably, which we are about to start in this lecture.

Human beings are able to process only about 16 different levels of brightness between black and white. A computer can handle any number of brightness levels when looking at the properties of a given pixel, right up to the set of values defined by the radiometric resolution of the sensor. Recall an 8-bit sensor can represent 256 levels of brightness per band.

An area which is easy for human beings to work with but less so for computer algorithms is the determination of shape, proximity and general spatial analysis. Humans are easily able to discern crop fields, roadways, lakes and other shapes, and their locations and juxtapositions, whereas quite complex software algorithms are required for shape and spatial analysis by machine.

A conclusion that can be drawn from this analysis is that the most successful image interpretation exercises in remote sensing are those in which the skills of the analyst are used to best effect with the powerful properties of computer algorithms.

Slide 1.15.4 As inferred in the last slide most image interpretation exercises in remote sensing will be based on machine classification (now often called *machine learning*) but will nevertheless involve the input of a skilled human analyst. We will always keep that in mind, even though this course will be heavily focussed on machine methods for image interpretation.

The idea is to use human knowledge optimally to get the best results out of the machine classification algorithms, in such a way that applying human knowledge on a small part of the data yields an understanding of the whole image in the labelling stage—usually a really good investment in time and money.

Slide 1.15.5 This slide represents the classification process in four blocks. The first is the potentially expensive one in which the analyst has to find some labelled training data to feed into the classifier algorithm. As noted here, in a real exercise this can be a time consuming and expensive step because it often involves field visits.

The second step is to use the labelled training data to "train" the particular classifier algorithm that the analyst has chosen to use. Software is always available for that task.

The third step is the important one and where we reap the benefits of classification. After having put a lot of effort into training the classifier, we now set it to work labelling the whole image.

The final step is to examine the thematic map produced by the classifier to see whether the errors made in the classification (and there will always be some) are within the bounds acceptable to the application at hand. Are we happy that certain crops are labelled within 95% accuracy or do we want to do better, for example? In practice we often find that we have to iterate or re-do parts of our classifications to get the levels of accuracy required.

Incidentally, how can we assess the accuracy of the thematic map? To do so requires further labelled samples, just like the samples we used for training. Indeed, during the acquisition of training pixels, the analyst will, at the same time, set aside some labelled pixels as "testing data" to be used when looking at how well the classifier has performed. We will consider that detail later in our course.

Slide 1.15.6 Here we have shown the two steps of training and labelling in a very simple classification exercise. The image is a segment of a four band Landsat multispectral scanner image with four predominant classes – vegetation, burned vegetation, urban and water.

Here it is easy to select training data from an inspection of the colour composite image product, as shown in the middle image. The training pixels of water are shown in purple, those for vegetation are shown as green, those for burned vegetation are in red and those for the urban regions in deep blue.

Those training pixels are fed to the classifier to produce the thematic map and table of areas on the right-hand side.

Slide 1.15.7 Much of our attention in image understanding will now focus on machine methods for analysis, based on training as just illustrated or, in some cases, using algorithms that discover the structure of the spectral space in which the regions are labelled afterwards, rather than during a training step.

Slide 1.15.8 We will look at a number of techniques; some are simple and easy to use, but other, more complex algorithms, often give better results. As seen in these illustrations, straight lines might be simple separating boundaries, but higher order curves can separate more interwoven data sets. In general, the latter are more difficult to implement than just placing straight lines between the classes.

Up to now we have talked about lines, planes and surfaces, which are concepts we normally associate with two- and three-dimensional drawings. Once the dimensionality exceeds three, we use the terms ***hyperplanes*** and ***hypersurfaces***.

Slide 1.15.9 To finish off this lecture we want to revisit the spectral space itself, since it is fundamental to the development of many machine learning methods.

In a later lecture we will also see how to transform the space into other, sometimes more effective, representations of our recorded remote sensing image data.

Slide 1.15.10 Recall that the spectral space is just a cartesian coordinate system in which pixels plot according to their brightness value in each band, or axis.

At this point it is of value to remind you that the next lecture will commence a detailed analysis which involves the properties of vectors and matrices. If you do not have that background and wish to follow the development, please consult some standard introductory treatments of that material. You could also consult Appendix C of J.A. Richards, *Remote Sensing Digital Image Analysis*, 5th Ed., Springer, Berlin, 2013, which develops the material in a form matched to how we will use it in remote sensing.

Slide 1.15.11 Each pixel point is represented by a column vector, which is given a single symbol in lower case, bold font. It will have as many entries as there are bands. For an N band sensor, the column vector will be N dimensional. By representing the sensor measurements in this form, we can use the whole field of vector and matrix algebra in developing machine learning tools.

Slide 1.15.12 This summary reminds us of two aspects of the lecture: the human/machine comparison and the importance of the spectral domain.

Slide 1.15.13 The first question in this quiz asks you to think about the size of the spectral space. What does that also tell you about its density for a typical image that might consist of say a million pixels?

Lecture 16. Correlation and covariance

Slide 1.16.1 We now start our mathematical journey by looking at some properties of groups of pixels in the spectral domain.

Slide 1.16.2 Here we assume you have a background in vectors and matrices. If not, we will summarise the key points geometrically and descriptively so that the main results should be clear.

We are going to look at three important spectral space concepts in this lecture—the mean vector, the covariance matrix and the concept of correlation.

Slide 1.16.3 Because we like to talk about where pixels are located in the spectral domain, it is useful to have measures of where they are most likely to be found and how they spread about that location.

We start with the concept of the mean vector, which is just the average or mean value of a group of pixel vectors as illustrated in this slide.

We denote the mean vector by the symbol **m*;*** it is calculated by taking the mathematical average of the *K* pixel vectors. Another term for average is expectation. Note we have introduced the expectation operator which is just the calculation of the average as shown.

Slide 1.16.4 Here we show a simple calculation of the mean of two pixel vectors. Note that the calculation is performed separately for each element of the pixel vector. Coming back to the concept of expectation, the position **m** is where we most expect to find a pixel from that group.

Slide 1.16.5 The mean vector tells us the average position of the group of pixels in the spectral domain. We would now like to describe how they spread about that mean position. That is a concept we are familiar with from statistics, where we use the concept of variance or standard deviation for that purpose.

Slide 1.16.6 Because we are dealing with data of any dimensionality, the simple concept of variance does not apply. But we can develop a multidimensional equivalent. This is called the covariance matrix and is again defined in terms of an expectation, using the formula shown on this slide. If you look carefully you can see that it is very similar in structure to the formula for variance in a one-dimensional case. Let's explore this further.

First, though, note that the denominator term in the average calculation is *K*-1 and not just *K*. That gives a better estimate.

Slide 1.16.7 First, we subtract the mean vector from each pixel vector, as shown in the centre of this slide, just like we take the difference from the mean in a one-dimensional variance calculation. We then need to square the mean difference, which is the role of the right-hand term, but it is turned into a row vector by the transpose operation shown. In the next slide we will see what that does to the product.

We then take the average or expected value over all the mean differences squared. The end result is a new construction called a matrix, represented by the upper-case **C.**

We have added the subscript x to the symbol for the covariance matrix, since it relates to calculation where the pixel vectors are described by x. Later we will encounter other coordinate systems and will therefore use other subscripts.

Slide 1.16.8 This slide shows the result of multiplying a column vector by a row vector, where the row vector appears on the right-hand side of the column vector. This gives a square array of numbers called a matrix, in which the elements are the result of the rules for multiplying vectors. The dimensions of matrix are equal to the number of bands in the image data.

A different result will be obtained if the order of the vectors is changed. We will see an example of that later in the course.

To see how this all works we will now consider a simple example.

Slide 1.16.9 Here we have a two-dimensional data set of points that are distributed around the space in a squashed circle arrangement. They would be the pixel brightness values of an image with just 6 pixels.

A set of hand calculations is shown, including the mean vector, the steps to calculating the covariance matrix, and the covariance matrix which results.

Note that the covariance matrix has zeros in the upper right and lower left positions, and only has non-zero values down the diagonal entries of the matrix. Those diagonal values are, respectively, the individual variances of the data points in the horizontal (abscissa) and vertical (ordinate) directions.

We have just talked about the diagonal of the matrix. That is the set of entries that run from the upper left-hand side to the lower right-hand side of a matrix. This particular covariance matrix is called diagonal because it has zero entries everywhere except down the diagonal.

Slide 1.16.10 This slide shows that the diagonal elements of the covariance matrix (and the correlation matrix to follow) relate to individual bands, whereas the off-diagonal elements describe the relationship between one band and another. Because of the zeros, in this case there is no relationship between the bands.

Slide 1.16.11 Now consider another two-dimensional data set; here the data points seem to be spread in an elliptical pattern at an angle to the axes. The hand calculations for this data set show that the covariance matrix is non-diagonal—i.e. it has non-zero entries everywhere.

At this stage it is important to define the correlation matrix, whose elements are computed from those of the covariance matrix as shown—they are the equivalent covariance matrix elements divided by the square root of the product of the two covariance elements on the diagonal, one on the same row and one on the same column as the entry under consideration.

Let's see what the correlation matrices look like for the two data sets …

Slide 1.16.12 Here are the two data sets compared, including their covariance and correlations matrices.

Note that the correlation matrices always have "1" for all of their diagonal entries. You can see why that is the case from the definition of the previous slide. It says that the pixel brightness values in a given band are fully correlated with themselves.

Secondly, for the left-hand data set there are zeros in the off-diagonal entries, implying that there is no correlation between the corresponding two bands of data. What does that mean? Put simply it means that knowing the brightness of a data point (pixel) in one band we cannot, with any degree of certainty, predict what its brightness is likely to be in the other band.

Slide 1.16.13 If we examine the right-hand data set, the off-diagonal terms of its correlation matrix are non-zero and imply there is about 76% correlation between the bands. In other words, if it is bright in one band, it is, with about 76% certainty, likely to be bright in the other band. That is because of the way the data is scattered—largely about a line at an angle to the data axes.

Slide 1.16.14 As a reminder, for the first data set the points do not distribute other than as parallel to the axes. There is no correlation between the bands. One cannot say that a pixel will be bright in one band if it is bright in the other.

Slide 1.16.15 This is an essential but simple dot point summary of the important elements from this lecture. Note the last, particularly, because it is essential to what we are going to do next.

Slide 1.16.16 Note that the last question relates the dimensionality of the two matrices (and the mean vector) to the number bands recorded by a particular sensor.

Lecture 17. The principal components transform

Slide 1.17.1 We now come to a particularly important concept that you will encounter time and time again in remote sensing image processing. We will see it has many uses, from classification and the general representation of images on a display system, through to highlighting changes that have occurred with time.

Slide 1.17.2 It is called the principal components transform (PCT), or principal components analysis (PCA). All remote sensing image processing software packages will include a module for computing the transform.

Depending on your background you may find some of the mathematics here a bit complicated. However, once we have been through that we will summarise the essential steps towards computing the transform. They are straightforward and easy to apply in practice.

Slide 1.17.3 When we apply the principal components transform, we generate a new set of bands with which to describe the image. The pixel brightness values in these new bands turn out to be weighted linear combinations of the original pixel brightness values.

There are a number of image transformations of this nature that can be encountered in image processing, including the Fourier transform, the wavelet transform and band arithmetic. We will only treat the principal components transform in these lectures and make some reference to band arithmetic (in which the elements of pixel vectors from two different images are added, subtracted or divided) from time to time.

Slide 1.17.4 To start us thinking about principal components, consider again the two data sets from the previous lecture. One has no correlation between its bands (axes); the other has high correlation.

Note the last comment on the slide: our correlation matrices will always be diagonal matrices of the same dimensionality as the number of bands recorded by the sensor, with unity for each diagonal element. Although not important here, that type of matrix is called an identity matrix.

Slide 1.17.5 We commence our development of the principal components transform by asking whether we can take a correlated data set and somehow transform it into a new set of coordinates in which the data exhibits no correlation.

Consider data that is scattered in the shape of the shaded ellipse shown in this slide. This is a bit like the correlated set from our two examples. In the original coordinates (i.e. as recorded by the sensor) the data is highly correlated. Bright pixels in one band tend to be bright in the other, etc.

But if we rotated our coordinates anti- or counter-clockwise we see that there is a rotation angle (shown here as the **y** coordinates) that corresponds to a coordinate system in which the pixels in the data set are uncorrelated.

In the language of vector and matrix analysis the new coordinates can be generated from the old by the equation shown. The values of the g_{ij} define the actual extent of the rotation. What we have to do is find those values so that the correct degree of rotation is given.

Slide 1.17.6 We can write the matrix equation in the symbolic shorthand notation shown on this slide. Whereas vectors are bold lower case, matrices are bold upper case. If you don't know how to multiply vectors and matrices please consult some standard treatments, or be prepared to follow the remainder of this lecture as best you can, even though it is addressed to the specific case of rotating axes.

Here is the really important point: we define the new axes as those in which the data shows no correlation or, in other words, in which the new covariance and correlation matrices are diagonal—i.e. they have zeros everywhere except down the diagonal entries.

Slide 1.17.7 Since we are interested in the covariance matrix, let's examine how it appears in the new rotated (**y**) coordinate system. This slide shows the standard definition for the covariance matrix that we met in the last lecture, but we have added the **y** subscripts to remind us we are dealing in that coordinate space.

The mean vector in the **y** coordinates is easily related to its value in the x coordinates via the transformation matrix **G**, as shown. We can substitute that in the formula for the covariance matrix to get the very important expression in blue on the bottom right hand side of this slide. To get to that point we had to use some properties of matrices and vectors. If you do not have that background just accept the last expression, noting that the transpose of a matrix flips it about the diagonal.

Slide 1.17.8 The equation we last derived expresses the covariance matrix in the new **y** coordinate system in terms of the covariance matrix of the original **x** coordinate system. The two are linked by the matrix **G** and its transpose.

Remember, when we started this analysis we had the objective that the data would be uncorrelated in the **y** coordinates—or in other words, the covariance matrix in the **y** coordinates would be diagonal. That constraint allows us to recognise this equation as the so-called diagonal form of the original covariance matrix. That tells us immediately what the elements of the matrix **G** are, and also gives us the diagonal entries of the **y** space covariance matrix, remembering that the off-diagonal entries are all zero.

Each matrix has a set of properties called its eigenvalues and eigenvectors: we will see how to get them in the next lecture. Usually, there are as many of each as the dimensionality of the matrix.

The eigenvalues are just scalar quantities (lambda) that can be rank ordered from largest to smallest as seen at the bottom of this slide. They are the diagonal entries of $\mathbf{C}_y$, rank ordered from largest to smallest.

The eigenvectors are a corresponding set of vectors. The matrix $\mathbf{G}^T$ consists of the set of eigenvectors, so that **G** is a transposed matrix of eigenvectors.

Slide 1.17.9 It turns out that the eigenvalues of the covariance matrix in the original set of bands (coordinates) are the variances of the brightness in the new axes. Remember, these axes are the brightness values of the image pixels in a new set of bands. Those new bands are called the principal components of the original image.

We generate the actual principal components pixel brightness values using the elements of the eigenvectors of the original covariance matrix.

In the next two lectures we will demonstrate how all that is done, using a model example and a real set of images.

Those examples will also answer the question as to why it is good to have low correlations among the bands of data.

Slide 1.17.10 This summary just reminds us that the steps in principal components analysis are simple, provided we have the image processing software available.

Slide 1.17.11 These questions require an understanding of how to multiply vectors and matrices. They provide useful guidance for material to come.

Lecture 18. The principal components transform: worked example

Slide 1.18.1 In this lecture we are going to do a set of hand calculations of a principal components transform. You will not have to do this in practice. The purpose of the exercise is to demonstrate the importance of the eigenvalues and eigenvectors of the covariance matrix, and how we should interpret their specific values.

Slide 1.18.2 Here we reiterate the important aspects of the principal components transformation. The equation at the top shows how we generate the principal components (the new bands **y**) from the original set of bands **x**. That relies on the elements of the transformation matrix **G**.

We get the elements of **G** by computing the eigenvectors of the original covariance matrix $\mathbf{C}_\mathbf{x}$. Remember **G** is the transpose of the matrix of column eigenvectors.

The covariance matrix of the transformed data, i.e. in the **y** coordinate space, is the diagonal matrix of the eigenvalues of $\mathbf{C}_\mathbf{x}$.

Slide 1.18.3 Remember there are four steps in obtaining the principal components, as we saw in the previous lecture. We will compute each of these steps in our example.

Once we have finished the example you should be quite familiar with how the principal components of an image come about.

Slide 1.18.4 We will do our calculations using the highly correlated data set from the previous lectures. Here we see the data distribution again, its covariance matrix and the equation we need to solve to find its eigenvalues and eigenvectors.

That equation is called the "characteristic equation", which is in the form of a determinant. Lambda is an unknown, which will be given in the solution to the equation. It has several values, which will be the eigenvalues we are looking for.

As we noted in the last lecture, the identity matrix is just a diagonal matrix with "ones" in the diagonal positions. When multiplied by the scalar lambda, it gives a matrix of zeros except with lambda in each diagonal position.

Slide 1.18.5 To solve the characteristic equation we substitute for the covariance matrix as shown here. When matrices are added or subtracted that is done element by element, also as seen here.

For this simple two-dimensional case, the determinant is evaluated using the expression in grey on the right-hand side of the slide. When we use that formula the quadratic equation in the unknown lambda results as shown. For greater than two dimensions the evaluation of the determinant is less straightforward, but software procedures are available for the purpose.

The characteristic equation has the two solutions shown. What do they represent?

Slide 1.18.6 The values of lambda are the eigenvalues of the covariance matrix of the data (pixels) in the original coordinate system—i.e. in the original set of bands.

We can therefore write down immediately the covariance matrix in the new principal components coordinates which, remember, is the diagonal matrix of eigenvalues. This tells us that the variance of the data in the first of the new principal coordinate axes is 2.67 and that in the second principal component is 0.33. Thus, we expect to see the data scattered predominantly along the first principal component, as expected from the diagram we used initially to start our thinking about the principal components transformation.

It is common to express the amount of the variance accounted for by each of the principal components. Here we see that the first component accounts for 89% of the scatter of the data.

Slide 1.18.7 Having found the eigenvalues, we can now proceed to find the eigenvectors. When we have done so we will then be able to form the transformation matrix **G**.

There are as many eigenvectors as there are eigenvalues—one corresponding to each.

Let's start with the largest eigenvalue, which we now call lambda_1. The corresponding eigenvalue is a solution to the vector equation shown on this slide. When we substitute in the values we know, we get a pair of simultaneous equations in the unknown components of the eigenvector.

Slide 1.18.8 Both equations are in fact the same and yield the single relationship between the two eigenvector components shown.

We now introduce a new bit of information—i.e. that the eigenvectors have unit magnitude, as seen in the middle of this slide. The magnitude of a vector is the square root of the sum of the squares of the elements; since that sum has to be unity, we don't need to worry about the square root. It is sufficient to say that the sum of the vector elements squared has to be unity.

When we substitute into that the solution to the eigenvector equation above, we now have unique values for the two components of the eigenvector, allowing the eigenvector to be written as shown in the second last equation on this slide.

If we go through the same process using the second eigenvalue, we get the last equation on the slide.

Slide 1.18.9 We now write the two eigenvectors side by side in matrix form and then transpose that matrix to give the required matrix which transforms the original data into the set of principal components.

The second half of this slide applies the transformation matrix to the data set we are working with, yielding the new brightness values for the pixels in the new **y** coordinates.

Slide 1.18.10 Here we see the data plotted in the new (principal components) coordinates. Even visually we can assess that the data shows no obvious correlation. Its maximum spread (variance) is in the first principal direction and the second largest spread is along the second principal axis. Note that the scales of the abscissa and ordinates are different here, which tends visually to mask the fact that the variance horizontally is much greater than the variance vertically.

If we were dealing with data of higher dimensionality that trend of decreasing spread would continue, with each subsequent component having progressively less variance.

We now need to see what all this means in the context of a real image example, which is the subject of the next lecture.

Slide 1.18.11 Here we re-iterate the essential steps in producing a principal components transformation.

Slide 1.18.12 The first two questions here are just to test your understanding of the meaning of the eigenvalues in principal components analysis.

The last question is particularly important and will arise time and again whenever you use principal components analysis. It is a common question in any situation where we seek to discard low variance components of a transformed data set, not just with PCAs but with other transforms as well that compress data variance into a small number of components.

Lecture 19. The principal components transform: a real example

Slide 1.19.1 We now wish to consolidate the lessons from the last couple of lectures by looking at a real example.

Slide 1.19.2 We will use the two data sets in this real example, both drawn from the same sensor.

The first (at the top) is a four band set where the bands are spaced over the spectral range. The first two are in the yellow-red part of the spectrum, the third is in the near infrared and the fourth is in the middle infrared. This data set has a low degree of correlation.

In the second data set (at the bottom) we have intentionally chosen four wavebands close together in the near infrared range. This data set has a high degree of correlation. You can tell that by inspection because the bands all look the same.

In each case we also show a colour composite image product created using the band/colour combinations shown under the colour product. The colour composite for the correlated data set is not very colourful, whereas that for the low correlation data set is colourful; the colour very much helps us pick out details among the cover types, as we have come to expect from colour image products.

Slide 1.19.3 Now let's examine the principal components for the first of those image sets—i.e. the one which shows low correlation. This slide shows the computed eigenvalues of the covariance matrix and the eigenvectors, here displayed by row so that we can see which eigenvector corresponds to which eigenvalue.

For all of these examples the images have been contrast enhanced, otherwise the later principal components might all appear black.

Note that even though the eigenvalues show rapidly decreasing variances, the components PC2, PC3 and PC4 still show a lot of detail, such that when the second and third principal components are used with the first to create a colour product a fairly colourful result occurs. Compared with the original colour composite of the scene there is a little more detail evident in the PC colour composite than that formed from the original bands.

Slide 1.19.4 Now look at the principal components of the second, highly correlated, data set.

Here we see a much more rapid drop off in the eigenvalues, telling us that most variance is in the first principal component, which is what we see when looking at the PC images. Again, the PC2, PC3 and PC4 images would be too dark to view if they were not individually contrast enhanced.

However, don't overlook the fact that, while their variances might be almost negligible, there can be significant detail in the later PC images. Look at the row of buildings in PC4, for example.

In this case the colour composite formed from the first three PCs shows significant detail not evident in the colour composite formed from the original bands. Note especially the lime green areas in the upper right-hand side of the image, and several of the buildings down the left-hand side.

Slide 1.19.5 Here we compare the two principal components sets on one slide. Note especially: the different drop off in the eigenvalues; the more apparent detail in the PCs in the top case, meaning the variance for that image is more shared among the PCs; compared with the bottom in which the variance is highly compressed towards the earlier PCs; note also how PC2 and PC4 in the second case seem to highlight particular features.

Slide 1.19.6 This is the same set of PC images as on the previous slide.

By looking at the colour composite comparison we note that there will generally be a more highly coloured product from the PCs if the original data set has low correlation among the bands.

Another thing to note in both cases is that colours appear in the PC colour composites that we almost never see in the colour composites formed from the original data—lime green, and bright purple are examples. That is a direct result of removing the correlations. We will explore that more later.

Slide 1.19.7 This slide puts it all together so you can see the comparison between the two examples directly.

The general conclusion we should draw is that if the bands of an image data set are not strongly correlated to start with then computing the principal components images will not achieve a great deal.

There will always be some visual effect, unless there was absolutely no correlation in the original data; but in general, the more correlation there is among the original bands the greater will be the effect of principal components analysis. This can definitely be seen in the second example. Computing PCs gives a much better colour result, for example, than with the original data.

This slide allows you to see the original data and the principal components for both examples. Note that you can get an idea of the correlation among the bands just by looking at them. In the first example, they look different from each other and thus exhibit low correlation. In the second example (third row) they almost all look the same. That is why the colour composite looks almost monotone. It is in cases like this that principal components has the greatest effect.

Slide 1.19.8 Two important points emerge in this summary. First, even though later principal components may contain very little variance they may, nevertheless, contain some interesting detail, detail that is quite different from the rest of the image. Secondly, we get much better use of colour with principal components imagery than with the original data.

Slide 1.19.9 These questions stress the importance of the actual values of the eigenvalues and eigenvectors of the covariance matrix.

Lecture 20. Applications of the principal components transform

Slide 1.20.1 We now want to examine a number of applications of principal components analysis to see how it is used in practice.

Slide 1.20.2 There are essentially four areas in remote sensing image processing where the principal components transform finds application.

The first is to produce colour imagery that is easier to interpret manually, particularly when spatial features are important.

The second is to compress imagery, in terms of data volume, to benefit efficient transmission and storage.

The third is as an aid in classification in which we try to reduce the number of features that define the data vectors that have to be labelled.

The fourth is in detecting changes between two images of the same region, taken at different times.

We will now look at each of those in turn.

Slide 1.20.3 Often in manual image interpretation—photointerpretation—we use the displayed colours, and our knowledge of spectral reflectance curves, to identify various regions on the earth's surface, giving them labels corresponding to the cover types of interest.

However, in some applications, the colours themselves are not so important as such, but rather they can be an aid to seeing spatial structure in an image. That is particularly the case in exploration geology, where the delineation of surficial lithological units is important, as is the identification of linear features (lineaments).

The vivid colours produced in principal components imagery, especially for highly correlated images, often allow structure to be seen that is otherwise very difficult to discern.

Slide 1.20.4 We will now show an example from geology. As is often the case, geologically significant features occur in barren regions for which the bands recorded by remote sensing instrumentation tend to be fairly similar and thus highly correlated, especially in the visible and near-infrared regions of the spectrum.

It is exactly this type of image that benefits from an application of principal components analysis.

Slide 1.20.5 The images down the left-hand side of this slide are the four Landsat MSS bands of a region called Andamooka near central Australia. The standard colour composite, formed by displaying the second near infrared band as red, the red band as green and the green band as blue, is shown on the upper right-hand side of the slide.

The second column of images on the left are the four principal components generated from the original bands. The colour composite formed from the principal components, shown on the bottom right-hand side of the slide, allows features and regions to be seen that can't be discerned easily on the original colour composite and is thus of particular value in region delineation in geology.

Slide 1.20.6 The next slide shows an example with a similar intent, although this time in an estuarine region in Australia's Northern Territory.

Slide 1.20.7 The top row of images are the six reflective bands of a Landsat Thematic Mapper image segment recorded in the north of Australia (the thermal band has been left out). Shown also is the colour composite formed from the standard mapping of near infrared to red, red to green and green to blue.

In the middle strip are the six principal components of the TM image segment.

Two different PC colour renditions are shown by choosing different sets of PCs with which to create the colour products. Look for features that are easier to see in the PC colour composites than in the standard colour composite formed from the original bands. For example, note the good differentiation of the water features in the second example.

Note also the discretisation noise in the later principal components. That occurs because the signal variance—i.e. the actual variance of the real information left over in the principal component—is so small that the actual radiometric resolution of the sensor is showing up as a limiting factor.

Slide 1.20.8 Now consider the use of PCs for data compression.

Because the lower PCs of an image often contain little variance—and, by implication, little information—they can often be discarded without sacrificing much useful information, on the average. Each PC that is discarded saves that much data volume for transmission or storage.

If the original image is to be reconstructed from the remaining PCs the inverse of the transformation matrix is used, with the discarded PCs replaced by matrices of zeros.

Slide 1.20.9 When we come to look at methodologies for using classification algorithms to produce thematic maps, we will often try to limit the number of bands or features used in the process. There are, as we will see, a number of good reasons for doing that, including the speed of the process.

The original number of bands as recorded define the features to be used in classification. In our classification methodologies we will look to reducing that number.

There are actually many ways of reducing the number of features in a classification exercise. The use of PCs is one of the simplest, which we now consider in outline.

Remember there are as many PCs as there are original bands. If we transform the data we can remove the least important PCs and thus have reduced data dimensionality to handle. In other words, the original feature vectors, which have as many elements as there are bands recorded, are replaced by vectors of PC brightness values, but with only as many elements as the number of PCs retained. In some cases, that can be a very large reduction in the dimensionality of the feature vectors, giving a major improvement in the efficiency of the classification process.

We will demonstrate this technique later in the course.

Slide 1.20.10 We now come to a very interesting application of principal components which had its origins in a 1979 conference paper (Byrne and Crapper, Landsat'79, Sydney, Australia, 1979) and has been "rediscovered" many times since. It relates to the use of PCs to detect features which have changed from image to image in a multi-temporal data set.

Slide 1.20.11 We will develop this application of PCs by considering a model flooding example, following which we will look at a real case involving bush fires.

Both examples involve two images of the same region on the earth's surface, which means the images have to be co-registered beforehand.

Slide 1.20.12 We will be processing a "before" and an "after" image together, and thus will be dealing with twice the number of bands as with a single image. While that will happen in our later real example, we will work our analysis through by just looking at a near infrared band from before the event of interest and the corresponding band from the image taken after the event of interest.

As will become clear, it is important, if the technique is to work well, that the number of pixels in the image which change as a result of the event of interest not make up more than about 10-20% of the scene. All the other pixels remain the same from image to image, apart from natural variations and possible seasonal differences, which give rise to changes in the level of solar irradiation.

Slide 1.20.13 This two-dimensional, multi-temporal spectral space allows us to see what happens with a dynamic event such as a flood. We have plotted the near infrared brightness values of the image pixels in a space with the IR response in the first date plotted horizontally and that for the second date plotted vertically.

Pixels which do not change from image to image, except for natural variations, scatter about a diagonal in the spectral domain, in a roughly ellipsoidal fashion. The angle which that scatter makes to the axes will be determined by the relative solar irradiation levels. If there are no seasonal differences, it will be close to 45 degrees in this two dimensional space.

Any pixels that were originally vegetation or soil, but have become flooded in the second date, will appear as a group in the region of space defined by high IR response in date 1 and low IR response in date 2.

Slide 1.20.14 We now envisage where the axes of the multi-date principal components transform will appear in this space. Recall that the first PC will be in the direction of maximum data variance and the second will be at right angles (orthogonal) to it and in the direction of the second most data variance, and so on in general. Also, the PCs are essentially just a rotation of the original axes in the anti- or counter-clockwise direction.

These axes are shown in red dashed form in this slide, overlaid on the original data set. In the second PC we see that the flooded pixels are readily separated from those which don't change substantially from date to date. This is emphasised on the next slide, shown separately for convenience.

We now see why it is important that the change pixels not occupy too much of the scene. We want the first axis to be set by the pixels that are essentially constant.

Slide 1.20.15 Here we see explicitly that in the second principal component the static cover types occupy a different range of values of PC2 than the range occupied by those cover types which change between images. Thus, they are easily separated and will appear with very different brightness vales when displayed. Here the flooded pixels would appear to be dark by comparison to those which are relatively constant between dates.

Note that negative PC brightness values can be generated. That is not a problem theoretically but, for display, all brightness values need to be positive. That can be handled by adding a constant to all the second PC values to make them all positive. That does not change the results.

Slide 1.20.16 Having established the basis of the technique—that the second and later PCs of a multi-temporal data set will show up isolated regions of change —we can now look at the real, bush fire example. This employs the same pair of images that we used in the image-to-image registration example of lecture 11.

What is appealing about this case is that there are fire related events in "both directions". The fire scar from 1979 is regenerating in the 1980 image while two new fire scars appear in 1980 that were not there in 1979.

Slide 1.20.17 As with the flood example this slide shows where the two PC axes would appear in the multi-date spectral domain, again highlighting that the major changes show up best in the second PC (in this two-dimensional illustration).

Slide 1.20.18 This slide reminds us that the exercise involves Landsat MSS data for which there are four bands in each date, so we are dealing with an 8 dimensional multi-temporal image and from which we compute 8 principal components.

When we look the at the results in detail we will see that the first PC looks like a weighted sum of all the 8 original bands, and is thus often referred to (not quite correctly) as a total brightness image.

Slide 1.20.19 Here are the eigenvalues and eigenvectors (by row) of the 8 dimensional multi-temporal image. Note the rapid drop off in the sizes of the eigenvalues and note that all the elements of the first eigenvector are positive—that's why the first PC is a weighted sum of the original set of bands. All the other eigenvectors have some negative elements, which is why they are good at highlighting changes that have occurred between the dates.

Slide 1.20.20 Here we see the first four principal components from the 1979/1980 image pair. We cannot see the change events in the first PC but they are very obvious in PC2, PC3 and PC4. We will see those changes more vividly if we form a colour image from those three PCs, as in the next slide.

Look carefully at the first PC. It seems to have suppressed the fire events and looks like a very good image, with clear topography.

Slide 1.20.21 Here we show the colour composite PC image created with the mapping shown: PC2 to red, PC3 to green and PC4 to blue.

Note that the 1979 fire scar, which is revegetating in 1980 shows up as a red to magenta colour, the two new fire scars show as lime green and two stream beds on the south east part of the image show as deep purple-blue, which in this case represents complete revegetation from the 1979 fire.

This example has served to highlight the usefulness of the principal components transformation for highlighting differences from image to image. You should be able to find many other examples in the literature.

Slide 1.20.22 Examine the first PCs in the examples of this lecture and see how, as “total brightness” images, they highlight topography, whereas topographic detail is missing in many of the PC images.

Slide 1.20.23 The questions in this quiz ask you to think about computing the covariance matrix over a sub-set of an image. That allows much greater flexibility in using PCs, especially in applications like change detection and, in some cases, for feature reduction prior to classification.